THE GALOIS-FIXED POINT ON AN M_{23} HURWITZ SCHEME: EXACT RECONSTRUCTION AND CHARACTERISTIC-23 GEOMETRY

FRANÇOIS LE LAY

ABSTRACT. Huang, Jackson, Lee, Poonen, Pries, and Zhang construct a regular degree-23 M_{23}-cover from a seven-element inner Nielsen class and identify one class fixed by the relevant arithmetic Galois action, which yields their cover over $\mathbf{Q}$. We determine exactly the corresponding finite étale Hurwitz scheme of degree 7 and use its characteristic-23 geometry to distinguish that class from the other six.

Over $K_0 = \mathbf{Q}(\sqrt{-23})$, the coordinate algebra of the finite inner Hurwitz scheme is

$$A_{\mathrm{H}} = K_0 \times L, \qquad [L : K_0] = 6.$$

The relative Galois closure of L/K_0 has group S_6. Exact canonical models and degree-23 maps over A_{H}, together with certified branch cycles, recover every Nielsen class and identify the degree-one factor with the distinguished cover.

At the prime above 23, the normalization of the integral Hurwitz scheme has three closed points. On each pointed closed fibre, the normalization parameter of a distinguished singular point has minimal polynomial

$$(u - 16)(u^2 + 1)(u^2 + u + 1) = 0$$

and constructs an idempotent e_{sing} separating the degree-one and degree-six components. Independently, a relative-transporter construction in M_{23} produces an augmentation $\epsilon(\Theta)$ with the same two values on the seven exact covers. The certified identification of those covers with the seven Nielsen classes therefore proves

$$\epsilon(\Theta) = e_{\mathrm{sing}}.$$

We also construct a generic cohomological correspondence whose mod-2 intersection number is $\epsilon(\Theta)$. A direct nearby-cycle calculation shows that ordinary duality retains the sheet labels, while a mod-4 audit shows that the coefficient Bockstein vanishes after the effective lifts are forgotten. We therefore retain the two effective graph cycles in the logarithmic normalization complex and construct its half-graded Clifford-volume line. Its generic parity is $1 + \epsilon(\Theta)$; its special fibre factors into the split-node parity 1 and the returned parity e_{sing}. This gives a second, branch-cycle-free proof of the displayed equality without identifying tagged and untagged nearby-cycle pairings.

Two further calculations describe the distinguished cover itself. We explicitly realize the degree-4 projection whose existence and minimality were proved by Huang et al., obtaining a primitive bidegree-$(23, 4)$ equation. In its branch fibre above $T = 0$, the seven unramified points form a Fano plane, the eight colliding pairs form an affine $\mathbf{F}_2^3$-torsor, and their arithmetic gives a $2^4{:}\,\mathrm{PSL}_3(2)$ field tower. These structures enter the effective logarithmic orientation line, but the branch fibre alone does not distinguish the fixed Hurwitz point.

As a separate application, every integer specialization $t \equiv 2 \pmod{319}$ of the minimal equation has Galois group M_{23}, and controlled ramification selects an infinite mutually independent subfamily. Exact SageMath, Singular, Magma, GAP, Arb, and PARI/GP certificates accompany the computations.

2020 *Mathematics Subject Classification.* Primary 12F12; Secondary 11G32, 14H30, 20B25, 11R32.

Key words and phrases. Mathieu group M_{23}, finite Hurwitz scheme, Galois-fixed point, characteristic-23 reduction, Fano plane, inverse Galois problem.

Contents

1. Introduction

The construction of Huang et al. begins with an inner Nielsen class of seven elements in M_{23}, with branch-cycle classes $(2A, 23A, 23B)$. One class is fixed by the arithmetic Galois action considered there and produces their regular cover over $\mathbf{Q}$; here regular means that $\mathbf{Q}$ is algebraically closed in the Galois closure over $\mathbf{Q}(T)$. The existence and rationality of this class are known. The question addressed here is why the corresponding point is also distinguished by the intrinsic characteristic-23 geometry of the moduli problem.

Our first main result determines the finite étale scheme and its coordinate algebra exactly. Put $K_0 = \mathbf{Q}(\sqrt{-23})$.

To fix the field of definition in the terminology, "Galois-fixed" in the title refers to the class fixed by the arithmetic action in [1]. After the ordered classes $23A, 23B$ are defined over K_0, the theorem below identifies the corresponding moduli point as the unique K_0-rational point of the inner Hurwitz scheme. Its descent to a cover over $\mathbf{Q}$ is the result of [1] recalled above.

Theorem 1.1 (Exact arithmetic of the finite inner Hurwitz scheme)**.** *Let $\mathcal{H}_{\mathrm{in}}$ be the reduced inner Hurwitz scheme of normalized three-point covers with branch-cycle classes $(2A, 23A, 23B)$. There is a degree-6 field extension L/K_0 and an isomorphism*

$$\mathcal{H}_{\mathrm{in}} \simeq \operatorname{Spec}(K_0 \times L).$$

The relative Galois closure of L/K_0 has group S_6. Over $\operatorname{Spec}(K_0 \times L)$ there is an explicit family of canonical genus-4 curves equipped with degree-23 maps; its seven geometric fibres exhaust the seven Nielsen classes, and each map has monodromy M_{23}.

Thus the distinguished geometric point is the unique K_0-rational point of $\mathcal{H}_{\mathrm{in}}$; equivalently, it is the degree-one open-and-closed subscheme in the canonical $1+6$ decomposition. There are then two independent ways to detect its complement. The finite-group side gives the augmentation $\epsilon(\Theta)$ of a relative-transporter element. The geometric side evaluates a normalization parameter at a distinguished singular point of each pointed characteristic-23 reduction and gives an idempotent e_{sing}. The central result identifies them in the exactly reconstructed coordinate algebra.

Theorem 1.2 (Agreement of the two component idempotents)**.** *Let e_{sing} be the idempotent defined by the singular-point coordinate in proposition 3.10, and let $\epsilon(\Theta)$ be the relative-transporter augmentation of Exact computation 3.11. Then*

$$\epsilon(\Theta) = e_{\mathrm{sing}}$$

in the coordinate algebra of $\mathcal{H}_{\mathrm{in}}$. Under the isomorphism $A_{\mathrm{H}} = K_0 \times L$, both sides equal the primitive idempotent $e_6 = (0, 1)$.

The characteristic-23 geometry behind the second idempotent uses a feature visible inside the distinguished cover. Let $X \to \mathbf{P}^1_T$ be the degree-23 cover constructed in [1]. Its splitting field over $\mathbf{Q}(T)$ is regular with Galois group M_{23}, its source curve has genus 4, and its branch points are

$$0, \qquad \sqrt{-23}, \qquad -\sqrt{-23}.$$

The inertia involution at $T = 0$ has cycle shape $1^7 2^8$: seven sheets remain fixed and eight pairs collide. These two sets support respectively a Fano plane and an affine $\mathbf{F}_2^3$-torsor. Their common odd fixed-point identity supplies the finite permutation sets in a second explanation of theorem 1.2. The ordinary nearby-cycle stalk keeps the individual labels and its coefficient Bockstein vanishes. In section 3.5 we instead retain the effective integral graph cycles, construct a half-graded determinant line on their logarithmic normalization complex, and use its specialization to obtain the comparison without the exact branch-cycle crosswalk.

Two distinct seven-element Galois sets must therefore be kept separate. The seven geometric points of $\mathcal{H}_{\mathrm{in}}$ parametrize seven covers. The seven unramified points above $T = 0$ lie inside only the distinguished cover. The second set contributes geometry to the quadratic-orientation explanation, but it is not itself the moduli scheme and does not by itself distinguish the fixed point.

To make the arithmetic within the distinguished cover exact, we also carry out the minimal-coordinate construction anticipated by Huang et al. Write $F(T, V) \in \mathbf{Z}[T, V]$ for their integral source equation.

Theorem 1.3 (Explicit realization of the minimal-degree projection)**.** *There are explicit polynomials $J_0, J_1 \in \mathbf{Z}[T, V]$ and a primitive irreducible polynomial $P \in \mathbf{Z}[T, W]$ such that*

$$\deg_W P = 23, \qquad \deg_T P = 4,$$

and, in $\mathbf{Q}(X)$,

$$W = \frac{J_0(T, V)}{J_1(T, V)}, \qquad P(T, W) = 0, \qquad \mathbf{Q}(T, W) = \mathbf{Q}(T, V).$$

Consequently P defines the same regular M_{23}-cover. The projection $X \to \mathbf{P}^1_W$ has degree 4, and therefore explicitly realizes the minimum over $\mathbf{Q}$ established in [1, Remark 3.4]*.*

The coordinate change is regular across the structured branch fibre and preserves both of its residue fields.

Theorem 1.4 (Transport of the fibre above $T = 0$)**.** *There are primitive irreducible polynomials $\widetilde{H}_7, \widetilde{R}_8 \in \mathbf{Z}[W]$ of degrees 7 and 8 such that*

$$F(0, V) = 23^4 H_7(V) R_8(V)^2,$$
$$P(0, W) = -23^3 \widetilde{H}_7(W) \widetilde{R}_8(W)^2.$$

Moreover

$$\gcd\big(J_1(0, V), H_7(V) R_8(V)\big) = 1,$$

and substitution $W = J_0(0, V)/J_1(0, V)$ induces isomorphisms

$$\mathbf{Q}[W]/(\widetilde{H}_7) \xrightarrow{\sim} \mathbf{Q}[V]/(H_7),$$
$$\mathbf{Q}[W]/(\widetilde{R}_8) \xrightarrow{\sim} \mathbf{Q}[V]/(R_8).$$

These isomorphisms give Galois-equivariant bijections between the respective sets of geometric roots.

Finally, the minimal equation has a separate global arithmetic application.

Theorem 1.5 (Uniform specialization)**.** *For every integer*

$$t \equiv 2 \pmod{319},$$

the primitive part of $P(t,W)$ has degree 23, and its splitting field over $\mathbf{Q}$ has Galois group M_{23} in its natural degree-23 action.

Here and throughout, the phrase *fibre above the rational branch point* means the fibre of $X \to \mathbf{P}^1_T$ above $0 \in \mathbf{P}^1(\mathbf{Q})$; it does not assert that its points are $\mathbf{Q}$-rational. This is a characteristic-zero branch fibre. We reserve *closed fibre* for a model over a local base and use *reduction* for passage modulo a prime.

The logical spine of the paper is

$$\text{degree-7 Hurwitz algebra} \longrightarrow \text{seven exact maps and certified branch cycles} \longrightarrow \epsilon(\Theta) = e_{\text{sing}}.$$

The pointed reduction at 23 independently constructs e_{sing} and suggests a more conceptual explanation of the same equality. Section 2 separates the proved comparison from that open geometric problem before the computational details. The minimal projection and branch-fibre arithmetic are supporting results on the distinguished cover, and the specialization theorem is an independent application.

1.1. **Prior work and contributions.** The regular M_{23}-cover, its source equation and branch cycles, the numerical computation of the seven complex covers, and the identification of the Galois-fixed Nielsen class are due to Huang et al. [1]. They also prove that the source curve has $\mathbf{Q}$-gonality 4 and observe that a degree-4 coordinate could replace their degree-8 coordinate, without computing such a function. The branch-fibre and Hurwitz-scheme calculations in this paper begin from their cover and its accompanying data [2].

The braid-orbit calculation behind the seven-element Nielsen class goes back to Häfner [3]. The numerical three-point-cover method used by Huang et al. is that of Klug–Musty–Schiavone–Voight [4], implemented in the Belyi-map database of Musty–Schiavone–Sijsling–Voight [5]. The point–line Gassmann pair and the flag description of transvections in $\mathrm{PSL}_3(2)$ are standard. Likewise, the abstract solvability of the split embedding problem with abelian kernel is not at issue here [17]; our calculation identifies the fields cut out by this particular branch fibre. Huang et al. already obtain abstractly an infinite mutually independent family of M_{23}-extensions [1, Corollary 1.4(d)], and general Hilbert–Grunwald methods provide arithmetic progressions with prescribed specialization behavior [15, 16].

Relative to this prior work, the central contributions are the exact $K_0 \times L$ Hurwitz algebra, its S_6 Galois closure and seven exact maps, the characteristic-23 singular-position idempotent, and its exact identification with the relative-transporter augmentation. We additionally construct an effective logarithmic quadratic-orientation line giving a branch-cycle-free comparison, after showing why ordinary nearby-cycle duality and a natural pinching shortcut cannot supply it. Supporting results give the explicit minimal-degree equation and ruling field, the arithmetic and obstruction theory of the distinguished branch fibre, and the all-members congruence class modulo 319 with controlled ramification. The detailed attribution, limitations, and data provenance are recorded in section 9.

2. The fixed-point mechanism

There are two primitive idempotents in

$$A_{\mathrm{H}} = K_0 \times L : \qquad e_1 = (1,0), \qquad e_6 = (0,1).$$

The theorem of the paper compares two independently defined functions on the seven geometric points. The first is the augmentation $\epsilon(\Theta)$ of a relative-transporter element in $\mathbf{F}_2[2^3{:}\,\mathrm{PSL}_3(2)]$. Its certified values are 0 on the distinguished Nielsen class and 1 on the other six. The second comes from the normalization of the integral Hurwitz scheme at the prime above 23. Evaluation at the distinguished singular point has polynomial

$$(u-16)(u^2+1)(u^2+u+1),$$

and the function taking the values $(0,1,1)$ on these three closed points is e_{sing}.

The exact reconstruction identifies the seven maps with the seven Nielsen classes. It follows that both functions are e_6, which proves theorem 1.2. This proof is arithmetic and fully certified. It is important not to describe it as a label-free geometric explanation: the certified branch-cycle matching is what places the finite-group values on the two factors of $K_0 \times L$.

There is nevertheless a concrete route toward such an explanation. For an involution $x \in 2A$, its seven fixed sheets form a Fano set H_x, and its eight two-sheet orbits form an affine $\mathbf{F}_2^3$-torsor A_x. Every element $c \in C_{M_{23}}(x)$ satisfies

$$|\operatorname{Fix}_{H_x}(c)| + |\operatorname{Fix}_{A_x}(c)| \equiv 1 \pmod 2.$$

Using this identity, finite incidence correspondences attached to the two wild endpoint normalizers realize $\epsilon(\Theta)$ as a genuine generic intersection number. This construction is given in section 3.5.

The local issue is subtle. A tempting Ferrand pushout identifies several generic labels at a closed point. Its nearby stalk at a quotient point q is not the one-dimensional constant sheaf: it is the full permutation module

$$\mathbf{F}_2[p^{-1}(q)].$$

Consequently the specialized pairing of the two lifted sections is still the coordinate pairing $\sum_e \alpha_e\beta_e$. The fibre-sum product $(\sum_e \alpha_e)(\sum_e \beta_e)$ appears only after an additional unit–counit operation and is a different cohomological correspondence. The two pairings already differ on a two-element fibre, so they cannot be identified by formal trace compatibility.

The exact local numbers point to the refinement used here. Before reduction modulo 2, the difference of the two branch coefficients is divisible by 2 but not by 4. Its half modulo 2 is the local value of a canonical half-weight quadratic refinement attached to the two effective graph cycles. The ordinary coefficient Bockstein nevertheless vanishes, both in the unrestricted graph-to-sheet cone and in the rank-one horizontal logarithmic nearby-cycle stalk. The construction in theorem 3.15 therefore retains the effective graph cycles as a distinguished Lagrangian in the logarithmic normalization complex and uses its half-graded determinant. This line carries the quadratic refinement without changing the ordinary nearby-cycle pairing.

3. Exact Hurwitz arithmetic and the component comparison

3.1. **Coordinate algebra and canonical models.** We now turn to the seven-cover moduli problem itself. The section has three tasks: reconstruct and identify the finite Hurwitz scheme, construct its component idempotents independently from finite-group and characteristic-23 data, and compare them in the exact coordinate algebra. The normalized canonical quadric has the normalization-invariant coefficient

$$J = \frac{q_{13}}{A^2},$$

where A is the coefficient used to impose the fixed normalization in the accompanying data. Computing J at the seven analytic classes and reconstructing over K_0 gives a finite étale K_0-algebra

$$A_{\mathrm{H}} = K_0 \times L, \qquad [L : K_0] = 6.$$

Let $\mathcal{H}_J = \operatorname{Spec} A_{\mathrm{H}}$. Its degree-one component carries the reconstructed map belonging to the distinguished class, and the six geometric points of its connected degree-six component carry the other six reconstructed maps. The branch-cycle computation below identifies $\mathcal{H}_J$ with the inner Hurwitz scheme $\mathcal{H}_{\text{in}}$.

Exact computation 3.1 (Coordinate algebra and canonical curves)**.** The linear factor of the polynomial of J is

$$J - \frac{148227 - 34830\sqrt{-23}}{142129},$$

and its complementary factor is irreducible of degree 6 over K_0. An absolute defining polynomial for L is

$$\begin{aligned} h(Z) = Z^{12} - 6Z^{11} + 20Z^{10} - 32Z^9 + 44Z^8 - 22Z^7 + 6Z^6 \\ - 22Z^5 + 44Z^4 - 32Z^3 + 20Z^2 - 6Z + 1. \end{aligned}$$

Over A_{H}, all ten coefficients of the canonical quadric, all twenty coefficients of a Petri cubic in the fixed coefficient normalization, and the two marked order-23 points reconstruct exactly. On each component the resulting quadric–cubic complete intersection is smooth.

The reciprocal form of h makes the Galois closure unexpectedly small to describe. Put

$$g(Y) = Y^6 - 6Y^5 + 14Y^4 - 2Y^3 - 27Y^2 + 44Y - 44.$$

Proposition 3.2 (Galois closure of the sextic component)**.** *Let $E = \mathbf{Q}[Y]/(g)$ and let $\widetilde{E}$ be its Galois closure. Then*

$$L = EK_0, \qquad \mathrm{Gal}(\widetilde{E}/\mathbf{Q}) \cong S_6, qquad\, \mathrm{Gal}(\widetilde{E}K_0/K_0) \cong S_6,$$

and

$$\mathrm{Gal}(\widetilde{E}K_0/\mathbf{Q}) \cong S_6 \times C_2.$$

The relative Galois group acts as the full symmetric group on the six remaining maps. In the embedding order of Exact computation 3.6, its letters have Nielsen labels

$$(7, 4, 1, 5, 3, 2).$$

Moreover

$$\mathrm{disc}(E) = 2^4 \cdot 11 \cdot 23^4, qquad\mathfrak{d}_{L/K_0} = (2^4 \cdot 11 \cdot 23).$$

Proof. Let a be the image of Z in $\mathbf{Q}[Z]/(h)$. The reciprocal-polynomial identity is

$$h(Z) = Z^6 g(Z + Z^{-1}).$$

The exact relative-to-absolute field map sends $\sqrt{-23}$ into $\mathbf{Q}(a)$, and the element $a + a^{-1} + \sqrt{-23}$ has degree 12 over $\mathbf{Q}$. Consequently $L = EK_0$. Maximal-order calculations give the displayed discriminants.

The rational primes $3, 139, 2671$ split in K_0, do not divide $\mathrm{disc}(g)$, and give respective factor-degree patterns

$$(6), \qquad (1, 5), \qquad (1, 1, 1, 1, 2)$$

for g modulo those primes. Thus the relative Galois group contains a 6-cycle, a 5-cycle, and a transposition. The first factorization also proves irreducibility. Of the sixteen transitive subgroups of S_6, only $6T16 = S_6$ contains all three cycle types.

Finally, the unique quadratic subfield of an S_6 splitting field is its discriminant field. Since

$$\mathrm{disc}(g) = 2^{22} \cdot 11 \cdot 23^4,$$

that field is $\mathbf{Q}(\sqrt{11})$, rather than $K_0 = \mathbf{Q}(\sqrt{-23})$. Therefore $\widetilde{E}$ and K_0 are linearly disjoint, giving the two asserted Galois groups. The labeling by Nielsen classes is obtained from the certified branch-cycle continuation. ∎

3.2. **Reconstruction and identification of the seven maps.** We now pass from the reconstructed coordinate algebra to the maps themselves. The analytic calculation supplies a uniform error bound; exact canonical-ring linear algebra then reconstructs the ratios, and certified branch cycles identify their moduli points.

Exact computation 3.3 (Uniform truncation-error certificate)**.** The 47 chart centres cover both half-triangles with pseudohyperbolic radius less than 0.471: an Arb calculation [10] on geodesically convex mesh cells in the Klein disk model gives the upper bound 0.4708683715. At Cauchy radius $\rho = 0.72$, the modes $0, \dots, 60$, augmented by four normalization rows, have certified minimum singular value at least $6.60 \cdot 10^{-4}$ in all seven classes. Let A be the augmented low-mode matrix, h the norm bound for the high-mode operator, and ℓ the bound for the low-mode transition operator. Parseval gives $h \leq 2.35 \cdot 10^{-11}$ and $\ell \leq 1.38 \cdot 10^3$, so the explicitly defined Schur-complement margin

$$m = \sigma_{\min}(A) - \frac{h\ell}{1-h}$$

is positive in every class.

A bootstrap using Cauchy estimates at outer radius 0.99 bounds the exact normalized forms. The final $N = 700$, $Q = 1280$ Acb calculation then evaluates all 1280 discrete Fourier transform (DFT) output modes, not only those through the polynomial cutoff. Combining its residual balls with the certified inverse and branch-jet conditioning gives at least 75 certified decimal digits in each of the first twenty branch-normalized Taylor rows used to recover the canonical quadric and Petri cubic.

The exact equations are too large to reproduce usefully in print. They are stored in power or integral bases in `hurwitz_canonical_models_candidate.json` and `hurwitz_marked_points_candidate.json`. Their importance here is that they allow the covering function to be recovered by linear algebra internal to the canonical ring, with no search for a low-degree plane ansatz.

Proposition 3.4 (Exact degree-23 ratios)**.** *Let $C \to \operatorname{Spec} A_{\mathrm{H}}$ be the reconstructed family of canonical complete intersections, and let $b, c : \operatorname{Spec} A_{\mathrm{H}} \to C$ be the two sections marking the points with order-23 inertia. There are explicit quintic canonical sections N, D such that*

$$u = \frac{N}{D}, \qquad \operatorname{div}(u) = 23b - 23c.$$

In particular u has degree 23 on every geometric fibre. Restriction to the degree-6 factor of A_{H} and its six K_0-embeddings gives six explicit conjugate maps.

Proof. The degree-5 piece of the canonical ring has dimension 27. The first 23 local coefficients at either marked point have rank 23, and hence

$$B = H^0(5K_C - 23b), \qquad C' = H^0(5K_C - 23c)$$

both have dimension 4. The degree-10 piece has dimension 57. Multiplication there turns the identities

$$B_i(MC')_j - B_j(MC')_i = 0 \qquad (0 \leq i < j \leq 3)$$

into a linear system with 16 unknowns. Its rank is 15; its kernel is generated by an invertible matrix M. The deterministic first row gives $N = B_0$ and $D = (MC')_0$. Exact local expansions give

$$\operatorname{ord}_b(N) = 23, \quad D(b) \neq 0, \qquad \operatorname{ord}_c(D) = 23, \quad N(c) \neq 0.$$

Finally, saturating the ideal generated by the four sections of B away from b gives the unit ideal, and the same calculation for MC' away from c again gives the unit ideal. Thus the two linear systems have no base points away from b and c. Together with the displayed local orders, this shows that N and D have no common zero and proves the divisor identity. $\square$

It remains to put the third branch point at 1. The raw value is $\lambda = u(a) \in L$, where a is any point above the order-2 branch point. Its coordinates in the chosen power basis have a large common denominator, so ordinary coordinatewise recognition is ineffective.

Exact computation 3.5 (Third branch normalization)**.** At 71 rational primes splitting completely in L, the unique nonzero linear factor of multiplicity 8 in the reduced branch discriminant is computed in all twelve residue embeddings and interpolated into the power basis. Their CRT modulus has 404 decimal digits. A 13-dimensional LLL calculation [9] gives a 1129-bit common-denominator vector; the next lattice vectors have at least 1246 bits. This value agrees with the independent 1024-bit Arb computation to $1.7 \cdot 10^{-101}$ in integral-basis coordinates. It also agrees exactly at 24 further completely split primes that were withheld from CRT and LLL.

For a characteristic-zero check, project the canonical complete intersection birationally to a plane sextic. After cancellation, N and D become plane forms of degree 7. The resultants of the plane sextic with a generic fibre and with $N - \lambda D$ both have degree 42. Their gcds with their x-derivatives have degrees 6 and 14, respectively; the common factor has degree 6, and the additional degree-8 quotient is squarefree. The only other possible source of such a repeated projected root is two distinct plane points with the same x-coordinate. Magma excludes this after good reduction at the completely split prime 863153: in residue embeddings $2, 3, 4, 5, 9, 12$, every irreducible factor of the degree-8 quotient gives a degree-1 gcd in the remaining coordinate. Since the resultant, gcd, and squarefree-factor degrees are preserved, a characteristic-zero common divisor of degree at least 2 would remain one after reduction. Thus the third fibre has eight distinct double points and seven simple points. Independently, the projective critical ideal has length $15 = 7 + 8$ in all twelve residue embeddings.

With this value, the six maps are normalized by

$$\beta = \frac{N}{\lambda D};$$

their branch values are $0, 1, \infty$, with passport

$$(23), \qquad (2^8 1^7), \qquad (23).$$

The exact field element λ, all reconstruction and holdout residues, and the sparse sections $N, D, \lambda D$ are stored in `hurwitz_degree23_branch_candidate.json` and `hurwitz_degree23_maps_candidate.json`. The degree and the two totally ramified fibres in proposition 3.4 are characteristic-zero exact statements, as is the third-fibre multiplicity in Exact computation 3.5. The uniform Taylor-tail estimate is supplied by Exact computation 3.3. The filenames retain the word *candidate* to record that LLL and bounded-denominator reconstruction were the discovery route for the displayed coefficients; the following independent cover identification removes that reconstruction assumption from the final Hurwitz-algebra statement.

Exact computation 3.6 (Exact eliminant and certified branch cycles)**.** In the affine canonical chart $X_0 = 1$, eliminate $X_3 = z$ from the quadric and Petri cubic. If their linear subresultant is $L_1(x, y)z + L_0(x, y)$, then every serialized quintic is linear in z, so a section $S_0 + zS_1$ becomes the degree-7 plane form $S_0L_1 - S_1L_0$. Eliminating y from the plane sextic and $N - t\lambda D$ gives a raw resultant $R_t(x)$ of degree 42. The exact gcd of R_2 and R_3 has degree 19 and divides every R_t; after division, the values at $t = 2, \ldots, 8$ interpolate

$$P(t, x) \in L[t, x], \qquad \deg_t P = 6, \quad \deg_x P = 23.$$

The unused value $t = 9$ agrees exactly. At $t = 0, 1, \infty$, the gcds with the x-derivative have degrees $22, 8, 22$, and the degree-8 gcd is squarefree.

For each of the six embeddings of L over the chosen embedding of K_0, Arb isolates the 23 roots over $t_* = 1/2 + 2i$ and continues them around rational polygonal loops about $0, 1, \infty$. On every straight path segment, each sheet is enclosed in a disk of radius R with fixed inverse slope Y. Writing

$$\eta = \sup |YP(t,c)|, \qquad q = \sup |1 - YP_x(t,x)|,$$

the segment is accepted only when $q < 1$ and $\eta + qR < R$, all 23 disks are pairwise disjoint, and the endpoint root balls lie in their unique disks. Otherwise the segment is bisected. Thus the continuation is certified by a uniform contraction, not by nearest-neighbour matching. The map on the degree-one component (Nielsen class 6) is checked independently from its integral minimal-degree equation by the same method.

The full run contains 150145 accepted tubes and has maximum dyadic depth 18. In the sextic-embedding order the Nielsen IDs are

$$(7, 4, 1, 5, 3, 2).$$

Together with the independently checked degree-one map, whose Nielsen ID is 6, the seven triples have cycle shapes

$$(23), \qquad (1^7 2^8), \qquad (23),$$

their products are the identity, and every pair generates a permutation group of order 10200960. After conjugating the order-23 generator to the fixed y, centralizer translation matches the seven involutions with Nielsen representatives $1, \ldots, 7$, exactly once each. The independent normalization of the degree-one map orders the two order-23 endpoints oppositely, so its comparison uses the inverse of its loop about 0.

Corollary 3.7 (Identification of the finite Hurwitz scheme)**.** *The finite étale scheme $\mathcal{H}_J = \mathrm{Spec}(K_0 \times L)$ is isomorphic to $\mathcal{H}_{\mathrm{in}}$. Thus $A_{\mathrm{H}} = K_0 \times L$ is the coordinate algebra of the finite inner Hurwitz scheme of degree 7 and type $(2A, 23A, 23B)$. In particular, all seven reconstructed maps have monodromy M_{23}.*

Proof. The group-side Nielsen calculation gives exactly seven inner classes. Exact computation 3.6 produces one exact cover in each of them. They are therefore distinct and exhaustive. The induced map between the two reduced finite étale K_0-schemes is a bijection on geometric points, hence an isomorphism. This identifies the coordinate algebra without an a priori height bound for the preceding LLL discovery. □

Proof of theorem 1.1. The coordinate algebra and smooth canonical curves are given by Exact computation 3.1. The exact maps are constructed in proposition 3.4 and Exact computation 3.5; their certified branch cycles give the identification with $\mathcal{H}_{\mathrm{in}}$ in Exact computation 3.6 and corollary 3.7. Finally, proposition 3.2 proves the S_6 assertion. □

3.3. **Local structure at 23.** The prime above 23 gives a particularly small local realization of this full symmetric action. Let $\mathfrak{q} = (\sqrt{-23})$ be the unique prime of K_0 above 23.

Proposition 3.8 (Local Hurwitz algebra at 23)**.** *The normalization of* $\mathrm{Spec}\, \mathcal{O}_{K_0,\mathfrak{q}}$ *in $A_{\mathrm{H}} = K_0 \times L$ has three closed points. Their ramification indices and residue degrees over $K_{0,\mathfrak{q}}$ are*

$$(e, f) = (1, 1), \qquad (1, 2), \qquad (2, 2),$$

where the first point lies on the degree-one component and the other two lie on the sextic component. In the S_6-closure, a decomposition group at $\mathfrak{q}$ is

$$D_{\mathfrak{q}} \cong V_4.$$

Its natural action on the six remaining points has orbits of sizes 2 and 4; inertia has cycle type $1^2 2^2$, and either lift of residue Frobenius has cycle type 2^3.

Proof. Modulo 23 the trace sextic satisfies

$$g(Y) = (Y-5)^2(Y+1)^4.$$

Over $\mathbf{Q}_{23}$ its two factors have degrees 2 and 4. Their discriminant valuations are 1 and 3, with residual units 9 and 7; their discriminant square classes are therefore 23 and -23. The quartic factor is totally tamely ramified. Its local Galois closure is dihedral of order 8: tame inertia is cyclic of order 4, while Frobenius acts by inversion because $23 \equiv -1 \pmod 4$. Its discriminant field is $\mathbf{Q}_{23}(\sqrt{-23})$. Restriction to $K_{0,\mathfrak{q}}$ gives the even Klein four subgroup. The quadratic factor field is $\mathbf{Q}_{23}(\sqrt{23})$; its compositum with $K_{0,\mathfrak{q}}$ is the same unramified quadratic field already contained in the dihedral closure. This gives the asserted group action and prime data.

The first residual equations make the common residue field explicit. If $s = \sqrt{-23}$, then

$$U = \frac{Y-5}{s}, \quad \overline{U}^2 = 15, \qquad V = \frac{(Y+1)^2}{s}, \quad \overline{V}^2 = 5.$$

Both equations define $\mathbf{F}_{23^2}$, since $15/5 = 3 = 7^2$ in $\mathbf{F}_{23}$. □

A complex Nielsen labeling of the local $2+4$ partition is not canonical: it requires choosing a prime of the S_6-closure above $\mathfrak{q}$, and changing that prime conjugates $D_{\mathfrak{q}}$ inside S_6.

We use the standard ADE nomenclature and the Milnor, Tjurina, and delta invariants as in [11].

Exact computation 3.9 (Pointed closed fibres at 23)**.** The exact canonical equations, marked points, and degree-23 map sections are integral at each of the three closed points of the normalized local Hurwitz scheme. The resulting closed fibres are geometrically integral quadric–cubic complete intersections, their canonical quadrics remain smooth, and the marked zero b and pole c remain distinct smooth points. Their singularities are

closed point	residue field	singularities
degree-one point (class 6)	$\mathbf{F}_{23}$	E_8
point with $(e,f) = (1,2)$	$\mathbf{F}_{23^2}$	E_8
point with $(e,f) = (2,2)$	$\mathbf{F}_{23^2}$	$A_2 + A_6$.

The Milnor and Tjurina numbers are respectively 8, and $2+6$. In each row the total delta invariant is 4, so the geometric normalization is $\mathbf{P}^1$.

Let N, D be the reduced quintic map sections. Their common base scheme has length 7, avoids the singular points, and

$$\operatorname{ord}_b(N) = \operatorname{ord}_c(D) = 23, \qquad \operatorname{ord}_b(D) = \operatorname{ord}_c(N) = 0.$$

Consequently, on the normalization,

$$\operatorname{div}(N/D) = 23b - 23c.$$

There is a unique coordinate t on the normalization with $b = 0$, $c = \infty$, and

$$\overline{\beta} = t^{23}$$

at all three closed points.

Proof. Exact reduction in the two residue fields gives the displayed singular loci and no singularity at infinity. Eliminating the smooth-quadric coordinate gives an irreducible plane sextic. The exact local plane equations have triple tangent, Milnor–Tjurina pair $(8,8)$, and fifth-order tangent restriction in the first two cases, identifying E_8 in characteristic 23. At the third point the two completed local equations have Milnor–Tjurina pairs $(2,2)$ and $(6,6)$ and critical-value orders 3 and 7, identifying A_2 and A_6. Their delta invariants exhaust the arithmetic genus 4. The common base ideal has constant Hilbert polynomial 7, and exact local series at b, c give the four displayed orders. Since the residue fields are perfect, the remaining scalar multiplying t^{23} can be absorbed into t. The resulting coordinate is unique, since $a^{23} = 1$ in characteristic 23 implies $a = 1$. □

These pointed closed fibres contain more arithmetic information than their ADE types alone. If P is one of the three closed points, let τ_P be the unique singularity in the E_8 cases and the uniquely characterized A_6 singularity in the third case. Both are unibranch. We may therefore evaluate the normalization coordinate at the point above τ_P.

Proposition 3.10 (Idempotent defined by the singular-point coordinate)**.** *Put $u(P) = t(\tau_P)$. On the reduced closed fibre of the normalized local Hurwitz scheme, u is a primitive element of its coordinate algebra, with minimal polynomial*

$$R_{23}(u) = (u - 16)(u^2 + 1)(u^2 + u + 1) = 0.$$

The linear factor is the degree-one E_8 point, the factor $u^2 + 1$ is the unramified quadratic E_8 point, and $u^2 + u + 1$ is the ramified $A_2 + A_6$ point. In

$$B = \mathbf{F}_{23}[u]/(R_{23})$$

the polynomial

$$e_{\text{sing}}(u) = 2u^4 + 2u^3 + 4u^2 + 2u + 3$$

is idempotent and has values $(0, 1, 1)$ on these three factors. Its unique lift to the completed local Hurwitz algebra is the sextic component idempotent $e_6 = (0, 1)$. In particular,

$$e_{\text{sing}}(P) = f(P) - 1 \pmod 2.$$

Proof. Exact evaluation on the normalizations gives

closed point	$u(P)$	minimal polynomial over $\mathbf{F}_{23}$
degree one, E_8	16	$u - 16$
$E_8,\ (e, f) = (1, 2)$	$\sqrt{-1}$	$u^2 + 1$
$A_6,\ (e, f) = (2, 2)$	r	$r^2 + r + 1.$

In the last row the A_2 point has parameter $t = 1$, while Frobenius exchanges r and r^2. The constructions of t and τ_P commute with residue-field automorphisms, so the displayed values define an element of the coordinate algebra of the reduced closed fibre. The three factors are pairwise coprime, and their degrees sum to the five geometric points of that fibre, proving the primitive-coordinate assertion.

If $A = (u^2 + 1)(u^2 + u + 1)$, then $A(16) = 11$ and $11^{-1} = 21$ in $\mathbf{F}_{23}$. Hence

$$e_{\text{sing}} = 1 - 21A = 1 + 2A,$$

which gives both its displayed formula and its three values. Idempotents lift uniquely through nilpotent thickenings and complete local rings. The two quadratic factors are precisely the two local factors of L, so the lift is e_6. □

Thus the characteristic-23 models themselves construct the morphism to the constant two-point scheme that cuts out the degree-six component; no complex Nielsen labels or continued branch cycles enter proposition 3.10. This construction alone does not yet identify the two expressions

$$\epsilon(\Theta) = e_{\text{sing}}.$$

The exact branch-cycle certificate proves the equality indirectly because both sides are e_6; this is the proof used in theorem 1.2. Section 3.5 then asks whether the equality can also be obtained from a single specialization argument. Resolving only the coarse ADE source is insufficient: the degree-one and unramified quadratic reductions both have geometric type E_8. A successful geometric proof must retain the arithmetic singular position together with compatible incidence and gluing data.

3.4. **Descent of finite invariants.** We first specify what *descent* means here. For a finite set S, let $\underline{S}_{K_0} = \coprod_{s\in S} \operatorname{Spec} K_0$ be the associated constant finite étale scheme. An assignment

$$f : \mathcal{H}_{\mathrm{in}}(\overline{K}_0) \longrightarrow S$$

descends over K_0 if it is invariant under $\mathrm{Gal}(\overline{K}_0/K_0)$, equivalently if it is induced by a K_0-morphism $\mathcal{H}_{\mathrm{in}} \to \underline{S}_{K_0}$. When $S = \{0,1\}$, such morphisms are equivalent to idempotents of the coordinate algebra A_{H}.

We next define the finite-group assignments to be tested. Let $G = M_{23}$, let (x, y, z) be one of the seven normalized triples, and put $Y = \langle y\rangle$ and $Z = \langle z\rangle$. The two-endpoint order count is

$$\nu(Y,Z) = \#\big\{(a,b) \in (\mathbf{F}_{23}^{\times})^2 : \mathrm{ord}(y^a z^b) = 23\big\}.$$

For $c \in \{y, z\}$ and $d \in \mathbf{F}_{23}^{\times}$, put

$$a_c(d) = \mathrm{ord}\big(x x^{c^d}\big), \qquad \kappa_m = \sum_{c\in\{y,z\}} \#\{d \in \mathbf{F}_{23}^{\times} : a_c(d) = m\} \quad (m = 2, 4).$$

Changing a generator of Y or Z only reindexes the exponents, and the sum over $c \in \{y, z\}$ is unchanged when the two endpoints are exchanged.

Write $N(Y) = N_G(Y)$, use the convention $x^g = g^{-1}xg$, and set

$$\Omega = x^{N(Y)} \cap x^{N(Z)}.$$

The stabilizer of x in either normalizer is trivial: its order divides both $|N_G(Y)| = 253$ and $|C_G(x)| = 2688$. Thus, for $w \in \Omega$, the transporter sets from x to w in $N(Y)$ and $N(Z)$ are singletons; denote their elements by $n_Y(w)$ and $n_Z(w)$. The relative transporter

$$c_w = n_Y(w) n_Z(w)^{-1}$$

centralizes x. Let A_x be the set of eight two-element orbits of $\langle x\rangle$ on the moved letters. The induced permutation $\bar{c}_w$ lies in

$$C_G(x)/\langle x\rangle \cong 2^3{:}\,\mathrm{PSL}_3(2) \cong \mathrm{AGL}(3,2),$$

acting faithfully on A_x. Define

$$\mu = \#\{\bar{c}_w : w \in \Omega\}, \qquad \Theta = \sum_{w\in\Omega} [\bar{c}_w] \in \mathbf{F}_2[C_G(x)/\langle x\rangle].$$

Finally, let ϵ be the standard augmentation homomorphism of this group algebra. Then $\epsilon(\Theta) = |\Omega| \bmod 2$. The Hamming weight $\mathrm{wt}(\Theta)$ is the number of nonzero coefficients of Θ in the group basis.

Exact computation 3.11 (Descent of the relative-transporter invariants)**.** In the sextic embedding order $(7, 4, 1, 5, 3, 2)$, the exact values of the two-endpoint order count ν, the affine-image count μ, and the Hamming weight of Θ are respectively

$$\nu|_L = (42, 46, 54, 46, 28, 54),$$
$$\mu|_L = (14, 28, 16, 28, 31, 16),$$
$$\mathrm{wt}(\Theta)|_L = (11, 27, 15, 27, 25, 15).$$

The cyclic-conjugacy order counts (κ_2, κ_4) are likewise

$$((4, 12), (2, 6), (0, 16), (2, 6), (0, 8), (0, 16)).$$

None is constant. By contrast,

$$\epsilon(\Theta)|_L = (1, 1, 1, 1, 1, 1), \qquad \epsilon(\Theta)|_{K_0} = 0.$$

Moreover, on all seven geometric points,

$$\mathbf{1}_{\{\nu=32\}} = \mathbf{1}_{\{\mu=17\}} = \mathbf{1}_{\{(\kappa_2,\kappa_4)=(0,12)\}} = \mathbf{1}_{\{\epsilon(\Theta)=0\}}.$$

Proposition 3.12 (Component idempotents detected by finite invariants)**.** *Let $e_1 = (1,0)$ and $e_6 = (0,1)$ be the primitive idempotents of $A_{\mathrm{H}} = K_0 \times L$. The four $\{0,1\}$-valued assignments in Exact computation 3.11 descend to morphisms to $\underline{\{0,1\}}_{K_0}$ and all correspond to e_1; the augmentation corresponds to $e_6 = 1 - e_1$.*

The assignments ν, μ, and (κ_2, κ_4) do not descend on $\operatorname{Spec} L$ *to morphisms into the corresponding constant finite schemes. Moreover Θ cannot descend as a section of a lisse $\mathbf{F}_2$-sheaf whose transition maps permute the natural group-algebra basis.*

Proof. Because L is a field, $\operatorname{Spec} L$ is connected. Every morphism from it to a constant finite étale scheme is therefore constant, whereas the displayed values of the three assignments vary among its six geometric points. A permutation of the group-algebra basis preserves Hamming weight, so the varying weights also rule out the stated basis-compatible descent of Θ. The four equality predicates are 1 only at ID 6, while the augmentation is 1 precisely on the other six points. Under the correspondence between morphisms to the constant two-point scheme and idempotents, they are therefore e_1 and e_6, respectively. □

This interpretation uses the certified $1 + 6$ decomposition and the exact branch-cycle identification of its seven geometric points.

Proof of theorem 1.2. By proposition 3.10, the singular-position function is $e_{\mathrm{sing}} = e_6$. By proposition 3.12, the augmentation is $\epsilon(\Theta) = e_6$. Hence $\epsilon(\Theta) = e_{\mathrm{sing}}$ in A_{H}. □

3.5. **The effective quadratic-orientation connector.** We first explain why ordinary nearby-cycle duality does not give the desired comparison, and then construct the additional orientation object that does. This gives a second proof of theorem 1.2; it is logically independent of the exact identification of the seven reconstructed branch cycles used in the first proof. Fix the natural degree-23 permutation representation $M_{23} \subset S_{23}$. Let $x \in 2A(M_{23})$, let $H_x = \mathrm{Fix}(x)$, and let A_x be the set of its eight two-element sheet orbits. Thus H_x is the seven-point Fano set and A_x is an affine torsor under $\mathbf{F}_2^3$. Put $B_x = H_x \amalg A_x$.

Lemma 3.13 (Fano–affine odd fixed-point lemma)**.** *For every $c \in C_{M_{23}}(x)$,*

$$|\operatorname{Fix}_{H_x}(c)| + |\operatorname{Fix}_{A_x}(c)| \equiv 1 \pmod 2.$$

Proof. Write $U = O_2(C_{M_{23}}(x))/\langle x \rangle \simeq \mathbf{F}_2^3$. The centralizer acts on H_x through $\mathrm{GL}(U)$ acting on $U^* \setminus \{0\}$ and on A_x through $U{:}\,\mathrm{GL}(U)$. Write the image of c as an affine map (t, g). If $g - 1$ is singular, its dual fixes a nonzero space of dimension d, so it fixes $2^d - 1$ points of H_x, an odd number. The affine fixed-point equation has either no solution or an affine space parallel to $\ker(g - 1)$, hence an even number of solutions when nonempty. If $g - 1$ is invertible, the dual action fixes no point of H_x, while the affine equation has exactly one solution. Both cases give the assertion. □

After a splitting extension, let $D_y, D_z \simeq 23{:}11$ be the two wild endpoint normalizers in one labeling. For $D = D_y$ or D_z, define

$$\mathcal{L}_D = \{(b, x^d, b^d) : b \in B_x,\ d \in D\}.$$

The action of D on x^D is free because a stabilizer order divides both 253 and $|C_{M_{23}}(x)| = 2688$. If $w \in x^{D_y} \cap x^{D_z}$, let $d_y(w)$ and $d_z(w)$ be the unique transporters carrying x to w. Then

$c_w = d_y(w)d_z(w)^{-1} \in C_{M_{23}}(x)$, and equality of the lifted labels over w is equivalent to $b^{c_w} = b$. The preceding lemma gives

$$\deg(\mathcal{L}_{D_y} \cap \mathcal{L}_{D_z}) \bmod 2 = |x^{D_y} \cap x^{D_z}| \bmod 2 = \epsilon(\Theta). \tag{1}$$

Equivalently, on the normalized finite branch-orbit scheme B and its graph-orbit incidence scheme I, take constant $\mathbf{F}_2$-sheaves and the two incidence correspondences $\mathcal{L}_{D_y}$ and $\mathcal{L}_{D_z}$ in opposite directions. Their Lefschetz–Verdier pairing on the generic fibre is exactly the finite intersection in (1). This names the schemes, sheaves, correspondences, and cohomological maps on the generic side.

3.5.1. *The local nearby-cycle calculation.* Let R be a strictly henselian discrete valuation ring with residue field k and fraction field K, and let $p : E \to Q$ be a finite surjection. The finite Ferrand pushout [18]

$$\mathcal{P}_p = \operatorname{Spec}(R^E) \amalg_{\operatorname{Spec}(k^E)} \operatorname{Spec}(k^Q)$$

has coordinate ring

$$A_p = R^E \underset{k^E}{\times} k^Q = \{(a_e) \in R^E : \bar{a}_e \text{ is constant on each fibre of } p\}.$$

For $q \in Q$, the strict local geometric generic fibre is the discrete set $E_q = p^{-1}(q)$. Therefore, with $\Lambda = \mathbf{F}_2$,

$$(R^0\Psi\Lambda)_q = \Lambda[E_q], \qquad (R^i\Psi\Lambda)_q = 0 \quad (i > 0). \tag{2}$$

Let $\alpha, \beta \in \Lambda^E$ select the two endpoint incidences. The generic cohomological maps are the identity on the selected normalization sections and zero on the others. Their generic pairing, and the pairing of their nearby-cycle specializations, are both

$$\sum_{e \in E_q} \alpha_e \beta_e. \tag{3}$$

Indeed, the specialized maps are

$$\Lambda \longrightarrow \Lambda[E_q], \quad 1 \longmapsto \sum_e \alpha_e[e], \qquad \Lambda[E_q] \longrightarrow \Lambda, \quad [e] \longmapsto \beta_e.$$

This is the elementary finite-stalk instance of compatibility of the Lefschetz–Verdier pairing with nearby cycles in Lu–Zheng [22, Corollary 3.10].

There are also canonical maps

$$\Lambda \longrightarrow \Lambda[E_q] \longrightarrow \Lambda, \qquad 1 \longmapsto \sum_e [e], \quad \sum_e a_e[e] \longmapsto \sum_e a_e.$$

Inserting these maps produces the fibre-sum, or untagged, product

$$\left(\sum_e \alpha_e\right)\left(\sum_e \beta_e\right). \tag{4}$$

It is a different correspondence, not the specialization of (3). For $E_q = \{1, 2\}$, $\alpha = (1, 0)$, and $\beta = (0, 1)$, the tagged pairing is 0 while the untagged product is 1. Thus a conductor quotient cannot by itself justify replacing (3) by (4).

The normalization–conductor triangle does not alter this conclusion. After restriction to the generic fibre its conductor-supported terms vanish; nearby cycles recover the full permutation module in (2), not a rank-one quotient. Applying the counit $p_*\Lambda_E \to \Lambda_Q$ is an additional operation and must be incorporated into the relative correspondence if it is to be used.

3.5.2. *The remaining quadratic-orientation problem.* There is a valid secondary calculation before reduction modulo 2. At a split tame node, the normalization quotient sends branch coefficients (a_+, a_-) to $a_+ - a_-$. When these coefficients agree modulo 2, their first mod-4 gluing obstruction is

$$\kappa(a_+, a_-) = \frac{a_+ - a_-}{2} \pmod 2.$$

For the augmented Mathieu packet the two exact coefficients are

$$a_+ = 1078, \qquad a_- = 112, \qquad \kappa = \frac{1078 - 112}{2} = 483 \equiv 1 \pmod 2.$$

This is a canonical divided gluing defect of the distinguished effective graph lift. It is not the ordinary nearby-cycle pairing of (1), and it is not an ordinary coefficient Bockstein after the effective lift has been forgotten.

Here is the obstruction explicitly. Put $\mathcal{C} = 2A(M_{23})$, let $T_D = x^D$ be a normalizer trace, and let $P_D = \mathcal{C} \setminus T_D$. For the natural sheet representation ρ, direct counting gives

$$\rho(\mathcal{C}) = 120J + 1035I, \qquad \rho(T_D) = 8J + 69I,$$

and hence

$$\rho(P_D) = 112J + 966I \equiv 2I \pmod 4.$$

The congruent lift $P_D - 2\mathcal{C}$, however, maps to $-128J - 1104I \equiv 0 \pmod 4$. Thus the coefficient Bockstein of the packet class in the unrestricted graph-to-sheet mapping cone is zero. The unit κ survives only while the two effective cycles $\mathcal{C}$ and T_D are retained separately.

Horizontal logarithmic nearby cycles do not remove this obstruction. Nakayama's formula [20, Theorem 3.5] gives, at a smooth point of one horizontal log divisor, one free coefficient copy in degrees zero and one. Reduction from $\mathbf{Z}/4$ to $\mathbf{F}_2$ is surjective in both degrees, so the coefficient connecting homomorphisms vanish. Split graph tags only form direct sums of these stalks.

The nonzero invariant has a quadratic description. On the even-weight subspace of a binary permutation module set

$$Q(v) = \frac{\mathrm{wt}(v)}{2} \pmod 2.$$

Then

$$Q(v + w) = Q(v) + Q(w) + \langle v, w \rangle.$$

In a common tag set containing the original and adjacent Mathieu $2A$-classes, one has

$$Q(T_{D_y} + T_{D_z}) = 1 + \epsilon(\Theta), \qquad Q(P_D) = 1.$$

On a pointed characteristic-23 branch, let $n \in N_{S_{23}}(D) = 23{:}22$ be the returned affine frame and let $q(n) \in \mathbf{F}_2$ record its coset modulo $D = 23{:}11$. The returned traces are equal when $q(n) = 0$ and lie in disjoint adjacent Mathieu classes when $q(n) = 1$. Consequently

$$Q(T_D + T_D^n) = q(n), \qquad Q(\mathcal{C} + T_D^n) = 1 + q(n).$$

The cross-term between P_D and $T_D + T_D^n$ is zero in both cases, so the last equality is exactly the polarization identity above with the finite gluing unit $Q(P_D) = \kappa = 1$. The pointed reduction computations give $q(n) = 0, 1, 1$ on the three closed characteristic-23 branches, which are the values of e_{sing}.

We now construct the line carrying this refinement. Let R be a ring in which 2 is invertible and let E be a finite free permutation module of even rank $2m$, with its standard orthonormal form. Define its half-graded determinant to be

$$\mathrm{hdet}(E) = \big(\det E, (-1)^m \det\langle\ ,\ \rangle_E, m \bmod 2\big). \tag{5}$$

The first two entries form a quadratic line and the last is its orientation parity. Formula (5) is the Clifford-volume formula: if $e_1, \ldots, e_{2m}$ is an orthonormal basis, then $(e_1 \cdots e_{2m})^2 = (-1)^m$. It is independent of the ordering and extends by alternating tensor product to based virtual perfect complexes of even Euler rank.

For two effective 0/1-cycles A, B in a common tag set, let e_A, e_B be their commuting idempotents. Then

$$s(A,B) = e_A + e_B - 2e_A e_B$$

is the idempotent of their symmetric difference. Put

$$\mathfrak{q}(A,B) = \operatorname{hdet}(\operatorname{im} s(A,B)). \tag{6}$$

Its parity is $Q(A+B)$. If S, R are even subsets, the based decomposition

$$\mathbf{Z}[S] \oplus \mathbf{Z}[R] \simeq \mathbf{Z}[S \triangle R] \oplus \mathbf{Z}[S \cap R]^{\oplus 2}$$

gives the line-valued polarization law

$$\operatorname{hdet}(\mathbf{Z}[S]) \otimes \operatorname{hdet}(\mathbf{Z}[R]) \simeq \operatorname{hdet}(\mathbf{Z}[S \triangle R]) \otimes \mathfrak{p}(S,R),$$

$$\mathfrak{p}(S,R) = \operatorname{hdet}\big(\mathbf{Z}[S \cap R]^{\oplus 2}\big), \qquad \deg \mathfrak{p}(S,R) = |S \cap R| \pmod 2. \tag{7}$$

Thus this is a quadratic Picard object, rather than an ordinary square root of a determinant line.

We shall use the following normalization statement. Its coefficient category is important: an object remembers the integral multiplicity of each normalized graph component, not only the resulting class modulo 2.

Lemma 3.14 (Effective normalization–gluing lemma)**.** *Let R be a ring in which 2 is invertible, and let X_0 be a split nodal curve whose dual graph Γ is a finite tree with a specified set of marked vertices containing every vertex of valence different from two. Fix two marked vertices, denoted v_- and v_+. At every unmarked vertex assume that the two restriction maps identify the same basis and that the decoration agrees under this identification. Let $\nu : \widetilde{X}_0 \to X_0$ be the normalization, and let $\mathcal{B}_{\mathbf{Z}}$ be a finite free integral based coefficient system on $\widetilde{X}_0$, with based identifications of its two branch stalks at every node; write $\mathcal{B} = R \otimes_{\mathbf{Z}} \mathcal{B}_{\mathbf{Z}}$ for its quadratic-line realization. For a branchwise effective integral coefficient z, put*

$$\delta_e(z) = z_{e,+} - z_{e,-}$$

after transporting both restrictions to the common basis at the node e. For an integral multiplicity vector $m = \sum_w n_w[w]$, write

$$R[m] = \sum_w n_w R[w]$$

in the Grothendieck group of finite based free R-modules; negative multiplicities are represented by dual factors. Thus z selects a virtual based object before any reduction modulo 2. Suppose that every coefficient of every $\delta_e(z)$ is even and that the unpaired virtual ranks are even, so their half-graded determinants are defined. Then the following assertions hold.

(1) *The classes*

$$\kappa_e(z) = \frac{\delta_e(z)}{2} \pmod 2 \quad \textit{in } \mathbf{F}_2[W_e]$$

are the node components of one divided normalization boundary. Along the unique path from the first marked vertex to the second, they satisfy the vector identity

$$\sum_e \kappa_e(z) = \frac{z_+ - z_-}{2} \pmod 2, \tag{8}$$

after all bases have been transported to one endpoint. Before reduction, the same identity gives, for the oriented path P,

$$\operatorname{hdet}\big(R[z_+] - R[z_-]\big) \simeq \bigotimes_{e \in P} \operatorname{hdet}\big(R[\delta_e(z)]\big), \tag{9}$$

with the tensor factors ordered along P.

(2) *The underlying normalization complex*

$$N^\bullet_\Gamma(\mathcal{B}) = \left[\bigoplus_v B_v \longrightarrow \bigoplus_e B_e\right], \qquad (db)_e = b_{e,+} - b_{e,-}, \tag{10}$$

is a finite based perfect complex. Apply the integral multiplicity vectors to its normalization filtration in the Grothendieck category of based perfect complexes. The determinant isomorphism then factors the half-graded determinant of this decorated virtual object into the marked vertex lines and the anomaly lines $\mathrm{hdet}(\delta_e(z))$*. Every unmarked internal based factor occurs once covariantly and once contravariantly and has its canonical evaluation trivialization.*

(3) *This factorization is functorial under simultaneous based relabelling, flat base change preserving the split normalization diagram, branch exchange, gluing of marked trees, and semistable subdivision. Branch exchange dualizes the corresponding anomaly line; subdivision inserts the based contractible complex* $[B \xrightarrow{1} B]$.

For the relative form, let R be a strictly henselian discrete valuation ring with $2 \in R^\times$, let A be a finite torsion-free graph-incidence R-algebra with finite normalization A^ν, and let z_η be an effective idempotent on $A^\nu \otimes_R K$. Assume that the conductor quotients are equipped with the based branch identifications of the split system above, that the resulting differences are coefficientwise divisible by 2, and that the unpaired virtual ranks are even. Then z_η extends uniquely to A^ν; its image is finite projective over R, and the conclusions above applied to the conductor complex are the generic and special factorizations of one relative quadratic line. The statement applies componentwise to a finite ordered family of idempotents; the line-valued polarization law then combines their factorizations.

Proof. At one split node, the normalization sequence with coefficients in the integral based module $B_{\mathbf{Z}}$ is

$$0 \longrightarrow B_{\mathbf{Z}} \longrightarrow B_{\mathbf{Z}} \oplus B_{\mathbf{Z}} \xrightarrow{(b_+,b_-)\mapsto b_+-b_-} B_{\mathbf{Z}} \longrightarrow 0. \tag{11}$$

The kernel of $\mathbf{Z}/4 \to \mathbf{F}_2$ is canonically $2\mathbf{F}_2$. Hence, if the branch difference of the distinguished integral coefficient is even, its reduction modulo 2 lies in the kernel of the branch-difference map and therefore descends across the node. Write $z_{e,+} - z_{e,-} = 2h_e$. Then $h_e \bmod 2 = \kappa_e(z)$ is precisely the cochain measuring the failure of the two branch lifts to agree modulo 4; equivalently, it is the first obstruction to descending the chosen integral lift modulo 4. This proves that the local classes are components of the normalization boundary. On an oriented path $v_0, e_1, v_1, \ldots, e_r, v_r$, their integral numerators telescope:

$$(z_1 - z_0) + (z_2 - z_1) + \cdots + (z_r - z_{r-1}) = z_r - z_0.$$

Dividing by 2 and reducing modulo 2 proves (8) as an equality of coefficient vectors, not merely of their augmentations. Before reduction, the same telescoping equality is an equality of virtual based modules. Applying the half-graded determinant in path order proves (9).

The local sequences (11) assemble to (10). Cutting Γ at the marked vertices decomposes it into chains whose internal vertices are all unmarked. Filter the complex on each chain by successively adjoining its vertices and edges. Each step is a short exact sequence of based complexes. Knudsen–Mumford determinant additivity therefore identifies its determinant with the ordered tensor product of the determinants of the graded pieces. These filtrations are based split: at every unmarked vertex the transition is a bijection of the distinguished bases. The Clifford-volume norm is multiplicative on such orthogonal based sums, with the doubled-overlap contribution recorded by $\mathfrak{p}$ in (7). At an unmarked internal vertex the same based module occurs in two consecutive normalization triangles, once in each dual cohomological position, so evaluation cancels the two factors. The unpaired pieces are the marked vertex terms and the virtual branch differences. Applying the Clifford-volume refinement to these same exact sequences gives the asserted half-graded factorization.

Gluing at marked vertices concatenates the two filtrations. If the common mark is forgotten after gluing and becomes an unmarked valence-two vertex, its two factors evaluate against each other; if the mark is retained, so is its factor. Subdivision adds $[B \xrightarrow{1} B]$, whose determinant has its canonical unit trivialization. Flat base change preserves (11); a based relabelling is an isometry; and exchanging the branches negates the virtual difference and therefore dualizes its line. This proves the functorialities.

For the relative assertion, z_η satisfies the monic equation $X^2 - X = 0$, so it belongs to the integral closure A^ν; uniqueness follows from the injection into the generic algebra. Its image is a direct summand of the finite torsion-free R-module A^ν, hence finite projective. Write $\mathfrak{c}$ for the conductor. The Ferrand square

$$A = A^\nu \times_{A^\nu/\mathfrak{c}} A/\mathfrak{c}$$

and its module sequence [18, Section 2.2] supply the two branch restriction maps to each common based conductor module. Their difference defines the relative complex (10); when the difference vanishes, its kernel is exactly the descended module, and in general its cone records the descent defect. The normalization terms are finite flat over the trait, while the conductor terms are finite over its discrete valuation ring and therefore have projective dimension at most one. The resulting complex is therefore perfect. Under a flat base change preserving this conductor diagram, derived base change and determinant base change identify its two fibre factorizations with those of one filtered relative complex and one determinant line. □

Theorem 3.15 (Effective logarithmic quadratic-orientation connector)**.** *On every completed pointed characteristic-*23 *Hurwitz branch, the effective graph-normalization complex carries a relative quadratic line $\mathfrak{O}_{\log}(\mathcal{C}, T_D)$. It retains the two effective cycles $\mathcal{C}$ and T_D separately and has the following properties.*

(1) *It is functorial under simultaneous graph relabelling; under flat base changes preserving the pointed split normalization diagram, including the finite splitting extensions used here; and under gluing and semistable subdivision of the marked component tree.*
(2) *On the generic fibre its orientation parity is*

$$\deg \mathfrak{O}_{\log,\eta} = Q(T_{D_y} + T_{D_z}) = 1 + \epsilon(\Theta).$$

(3) *The normalization–conductor filtration identifies its special-fibre parity with*

$$\deg \mathfrak{O}_{\log,s} = Q(P_D) + Q(T_D + T_D^n) = 1 + q(n) = 1 + e_{\text{sing}}.$$

The first summand is the horizontal split-node orientation and the second is the returned wild orientation. All unmarked valence-two transport factors occur in dual cohomological degrees and cancel.

Proof. Let S be the completed pointed trait. Over the étale locus of the marked Galois cover, its self-fibre product is the disjoint union of the deck-transformation graphs. Its normalized graph algebra is therefore the split based algebra

$$\mathcal{A}^\nu \simeq \prod_{g \in M_{23}} \mathcal{O}_{\Gamma_g}.$$

The characteristic functions of T_{D_y}, T_{D_z}, $\mathcal{C}$, and T_D are literal integral idempotents in this algebra. Take the closures of the graph components, pull them back to the marked finite branch-orbit scheme, and only then push along the natural degree-23 quotient. The resulting incidence algebra and its conductor are finite over S. Additivity of this pushforward sends an effective graph sum to its branchwise integral multiplicity vector. Thus the generic trace idempotents and the special coefficients used below are two restrictions of the same decorated graph-normalization datum; no Chow-theoretic refined-intersection identification is being inserted.

After a splitting base change, the graph components form a finite based coefficient system on the marked component tree joining the finite end and the two wild ends. Put

$$P_{D_y} = \mathcal{C} - T_{D_y}, \qquad P_{D_z} = \mathcal{C} - T_{D_z}.$$

These are effective idempotents because the traces are subsets of $\mathcal{C}$, and

$$s(P_{D_y}, P_{D_z}) = s(T_{D_y}, T_{D_z}).$$

We nevertheless retain the ordered pair (P_{D_y}, P_{D_z}), rather than collapse it to the common symmetric-difference idempotent. Each packet idempotent extends uniquely to the normalization of the finite incidence algebra: an idempotent is integral, and the normalization separates the generic graph components. Their image modules are finite projective over the splitting trait.

At the horizontal modification the decoration retains the two effective integral cycles $\mathcal{C}$ and T_D separately, rather than only their common reduction. Concretely, the generic and special unpaired graded presentations of this ordered pair are

$$\begin{aligned} \mathfrak{O}_\eta &= \mathfrak{q}(P_{D_y}, P_{D_z}) = \mathfrak{q}(T_{D_y}, T_{D_z}), \\ \mathfrak{O}_s &= \mathrm{hdet}(P_D) \otimes \mathfrak{q}(T_D, T_D^n) \quad \text{up to the polarization line.} \end{aligned}$$

These are two filtrations of the same relative virtual object, not standalone modules of the same rank. Between the marked charts the packet idempotents are carried by the based annular identifications. They need not descend separately as idempotents of the unnormalized graph algebra; their conductor boundary is precisely the additional datum retained by the decorated complex.

For a logarithmic node e, restrict these effective cycles separately to the two branches of the graph normalization and then push them to the sheet correspondence. Let $B_{e,+}$ and $B_{e,-}$ denote the resulting integral based incidence coefficients, and use the same notation for their associated virtual based modules. The coefficient vector of their difference is divisible by 2; this is the correct assertion, not an isomorphism between modules of unequal ranks. Hence

$$\mathcal{D}_e = B_{e,+} - B_{e,-}$$

has even virtual rank and defines the anomaly line $\mathfrak{a}_e = \mathrm{hdet}(\mathcal{D}_e)$.

The ordered idempotents also specify which quadratic line is being transported. In the normalization–conductor filtration, expand the first packet as $P_D = \mathcal{C} - T_{D_y}$ before cancelling the two copies of $\mathcal{C}$; its unpaired finite term is the effective packet P_D. The second trace is carried through the wild annuli and returns as T_D^n. Thus the special presentation is the terminal pair $(\mathcal{C}, T_D^n)$. Since $\mathcal{C} = P_D + T_D$ in the binary tag module, the line-valued polarization (7) factors its orientation into the packet line and the returned line. The polarization factor has degree

$$\langle P_D, T_D + T_D^n \rangle = 0,$$

in both return cases.

Ferrand's conductor square supplies the branch restriction maps defining the relative normalization complex $\mathcal{N}^\bullet$ of the two packet image modules with their effective decoration. Its generic fibre is the selected graph-label complex, and its special fibre has the cellular presentation

$$N_s^\bullet = \left[\bigoplus_v B_v \longrightarrow \bigoplus_e B_e \right], \tag{12}$$

with branch-difference differential and relative boundary conditions at the marked vertices. The hyperbolic completion

$$\mathcal{N}^\bullet \oplus R\mathrm{Hom}(\mathcal{N}^\bullet, R)[1]$$

is 1-shifted self-dual under the evaluation pairing. The ordered packet image modules in $\mathcal{N}^\bullet$, with their effective integral decorations, form the distinguished Lagrangian in this hyperbolic completion. Define $\mathfrak{O}_{\log}$ as the half-graded determinant of that Lagrangian.

The hypotheses and all the gluing assertions of lemma 3.14 now apply. It remains only to identify the unpaired local terms. On the finite one-node chart $t = u^2$, the completed self-fibre product is

$$A = R[[u, v]]/\big((u - v)(u + v)\big),$$

with normalization $R[[u]] \oplus R[[u]]$. The exact coefficient sequence is

$$0 \longrightarrow A \longrightarrow R[[u]] \oplus R[[u]] \xrightarrow{(f_+, f_-) \mapsto f_+(0) - f_-(0)} R \longrightarrow 0.$$

Thus its coefficient complex is literally the branch-difference complex

$$[\, B_+ \oplus B_- \longrightarrow B_{\mathrm{cond}}\,].$$

Restriction of an effective graph sum to the two normalized branches and additive quotient pushforward commute, so applying this sequence to $\mathcal{C}$ and T_D separately gives $\mathcal{D}_{\mathrm{fin}}$ without a further comparison theorem.

On every wild annulus, Wewers's patching datum [21, Section 4.2] is a based isomorphism of the effective graph modules. More precisely, one global patching datum is $(\lambda; \varphi_j)$, with the same group isomorphism λ for every tail. Since a G-equivariant φ_j carries Γ_g to $\Gamma_{\lambda(g)}$, every internal annular normalization cone is based acyclic. The only unpaired wild datum is the mismatch between T_D and T_D^n at the marked returned end.

The gluing lemma and its determinant filtration now identify the generic unpaired term with the symmetric-difference module of T_{D_y} and T_{D_z}. On the special fibre the unpaired terms are the finite divided-conductor line and the returned symmetric-difference module. Formula (7) combines their parities as the orientation of $(\mathcal{C}, T_D^n)$; its polarization factor has degree zero by the displayed cross-term calculation. All unmarked transport factors, gluing maps, and subdivisions are handled by lemma 3.14.

At the finite split node, the exact effective branch ranks are

$$\operatorname{rank} B_+ = 1078, \qquad \operatorname{rank} B_- = 112.$$

Consequently

$$\deg \mathfrak{a}_{\mathrm{fin}} = \frac{1078 - 112}{2} = 483 \equiv 1 \pmod 2.$$

The effective packet $P_D = \mathcal{C} - T_D$ has $|P_D|/2 = 3542/2 = 1771 \equiv 1 \pmod 2$, the same parity. More precisely, the divided normalization coefficient consists of the eight simple-node coordinates together with the fixed-heptad residual class. The intrinsic degree-three Fano incidence pairs the heptad with the seven nonfixed node coordinates; their binary contributions cancel, leaving the fixed-node line once. This is the finite unpaired term in the gluing lemma. At every other internal node, Wewers's common group identification gives a based isomorphism $B_{e,+} \simeq B_{e,-}$, so $\mathcal{D}_e = 0$ as a virtual based module. The nontrivial affine-normalizer mismatch is retained at the marked returned end, not as an additional internal-node anomaly.

Both generic traces have size 253, so

$$\deg \mathfrak{q}(T_{D_y}, T_{D_z}) = 253 - |T_{D_y} \cap T_{D_z}| = 1 + \epsilon(\Theta).$$

At the returned end the traces are equal when $q(n) = 0$ and are disjoint in the adjacent Mathieu copy when $q(n) = 1$. Their symmetric-difference ranks are 0 and 506, giving orientation parities 0 and $253 \equiv 1$. This is $q(n)$. The filtered determinant identity now gives the two fibre formulas in the statement. Since the grading of a relative quadratic line is locally constant on the trait, the two displayed parities are equal.

Finally, a change of splitting or of Wewers patching datum relabels every graph by the same group isomorphism. It is therefore an isometry of the entire decorated normalization complex, so the line descends from the splitting trait. The remaining functorialities are precisely those of lemma 3.14. □

Corollary 3.16 (Branch-cycle-free comparison)**.** *The effective logarithmic orientation line gives*

$$1 + \epsilon(\Theta) = 1 + e_{\mathrm{sing}},$$

and hence a second proof of $\epsilon(\Theta) = e_{\mathrm{sing}}$ *that does not use the certified identification of the seven characteristic-zero branch cycles.*

Ordinary nearby-cycle duality alone cannot produce this line. It forgets the distinguished Lagrangian and identifies the effective packet with the congruent lift $P_D - 2\mathcal{C}$; the latter has zero mod-4 sheet operator. By contrast, changing the effective lift by $-2\mathcal{C}$ changes the half-determinant parity by $|\mathcal{C}| = 3795 \equiv 1 \pmod 2$. Thus dependence on the ordered effective cycles is a necessary feature, not a choice of presentation. No result of Lu–Zheng is used to identify tagged and untagged pairings or to turn the divided node class into a coefficient Bockstein.

The next two sections record supporting arithmetic on the distinguished cover: first its explicit minimal projection, then the Fano and affine arithmetic of its branch fibre.

4. Supporting arithmetic: the minimal-degree projection

This section makes the distinguished cover explicit in the smallest $\mathbf{Q}$-defined source coordinate. Its role in the main argument is supporting rather than selective: the coordinate transports the Fano and affine residue data exactly, but it does not distinguish one point of the seven-cover Hurwitz scheme.

Set $D = T^2 + 23$. The coefficient of V^{23} in F is D^4, so

$$\Phi := F, \qquad \widehat{F} := D^{-4}F \in \mathbf{Q}(T)[V]$$

defines the same function field and is monic of degree 23 in V. We write C_0 for the affine plane curve $\Phi = 0$ and X for its normalization. The compactification under consideration is the closure in $\mathbf{P}^1_T \times \mathbf{P}^1_V$.

4.1. **The two points above $T^2 + 23 = 0$.** Let $b, c \in X(\overline{\mathbf{Q}})$ be the unique points over the roots of D. They are conjugate over $\mathbf{Q}$ and each has ramification index 23 for the T-map [1, Section 3].

Lemma 4.1 (Valuations at the two ramification points)**.** *At either* $q \in \{b, c\}$ *one has*

$$\mathrm{ord}_q(T - T(q)) = 23, \qquad \mathrm{ord}_q(D) = 23, \qquad \mathrm{ord}_q(V) = -4,$$

and

$$\mathrm{ord}_q\left(\frac{dT}{\Phi_V}\right) = 18.$$

Proof. The first two equalities follow from total ramification and the fact that D has a simple zero at $T(q)$. The coefficients of Φ, viewed as a polynomial in V, have a Newton-polygon edge joining $(0,0)$ and $(23,4)$ at $D = 0$. Hence its unique branch has $\mathrm{ord}_q(V) = -4$.

For completeness, the terms of Φ_V on this branch have orders

$$23\,\mathrm{ord}_D(ia_i) - 4(i-1),$$

where $a_i(T)$ is the coefficient of V^i. The minimum is 4, attained uniquely by differentiating the D^4V^{23} term; all other orders are at least 5. Thus $\mathrm{ord}_q(\Phi_V) = 4$. Finally $\mathrm{ord}_q(dT) = 23 - 1 = 22$, so $\mathrm{ord}_q(dT/\Phi_V) = 22 - 4 = 18$. □

The calculation is reproduced directly from the normalized coefficient table by `verification/verify_boundary_valuations.gp`.

For a polynomial $A(T,V)$ consider the adjoint differential

$$\omega_A = \frac{A(T,V)\,dT}{\Phi_V(T,V)}.$$

The interior of the bidegree-$(8,23)$ rectangle gives the usual support $0 \le i \le 6$, $0 \le j \le 21$ for monomials T^iV^j. Regularity at b and c sharpens this $7 \cdot 22 = 154$-dimensional space substantially.

Lemma 4.2 (Adjoints regular at b and c)**.** *The differentials ω_A having interior support and regular at both b and c are represented by the 88-dimensional space*

$$\mathcal{A} = \bigoplus_{j=0}^{4} V^j \mathbf{Q}[T]_{\le 6} \oplus \bigoplus_{j=5}^{10} DV^j \mathbf{Q}[T]_{\le 4} \oplus \bigoplus_{j=11}^{16} D^2V^j \mathbf{Q}[T]_{\le 2} \oplus \bigoplus_{j=17}^{21} D^3V^j \mathbf{Q}.$$

In particular,

$$\dim \mathcal{A} = 35 + 30 + 18 + 5 = 88.$$

Proof. Write $A = \sum_{j=0}^{21} a_j(T)V^j$. By lemma 4.1, a term $D^m c(T)V^j\,dT/\Phi_V$ has order

$$23m - 4j + 18$$

at b and c, unless c also vanishes there. Regularity therefore requires

$$m \ge \left\lceil \frac{4j-18}{23} \right\rceil,$$

giving respectively $m = 0, 1, 2, 3$ on the four displayed blocks. Terms having distinct exponents j cannot cancel: their orders are distinct modulo 23. Divisibility by D^m lowers the residual T-degree from 6 to $6 - 2m$. Counting coefficients gives the stated dimension. □

4.2. **The affine nodes and the canonical system.** At an ordinary node, Rosenlicht's adjoint condition [7] is the vanishing of the numerator A. Thus each node contributes one linear evaluation condition, and the rank computation below verifies independence. We impose the conditions in one stroke using the reduced singular scheme modulo 31.

Exact computation 4.3 (The node graph modulo 31)**.** Over $\mathbf{F}_{31}$ the Jacobian ideal

$$(\Phi, \Phi_T, \Phi_V)$$

has lexicographic basis

$$h_{84}(T), \qquad V - s(T),$$

where $\deg h_{84} = 84$ and h_{84} is squarefree. The determinant of the Hessian is a unit on this scheme. Thus the affine singular scheme consists geometrically of 84 reduced ordinary nodes.

This statement is certified by `verification/verify_nodes_mod31.sing`; the explicit h_{84} and s are in `data/node_graph_mod31.inc`. Evaluation of the basis in lemma 4.2 on the node graph gives an 84×88 matrix over $\mathbf{F}_{31}$.

Exact computation 4.4 (Canonical kernel and degree-4 pencil)**.** The node-evaluation matrix has rank 84, so its kernel has dimension 4. This is the canonical adjoint space $H^0(X, K_X)$. On this kernel, vanishing at $b + c$ has rank 2, leaving the two-dimensional space $H^0(X, K_X - b - c)$ and hence a pencil.

Explanation of the two conditions at b and c. Among the blocks of lemma 4.2, the only term that can have order zero at b or c is

$$D^2(c_0 + c_1T + c_2T^2)V^{16}.$$

Its vanishing at both roots of D is equivalent to divisibility of the quadratic coefficient by D, namely

$$c_1 = 0, \qquad c_0 = 23c_2.$$

These two conditions have rank 2 on the four-dimensional node kernel. The ranks are reproduced by `verification/verify_adjoint_mod31.py`. □

The successive dimensions are therefore

$$154 \longrightarrow 88 \longrightarrow 4 \longrightarrow 2.$$

The last two arrows are certified by full-rank minors modulo 31. This good-reduction calculation is the modular construction step; it does not by itself certify the characteristic-zero equation. That role is played by the exact function-field identity and irreducibility proof in section 4.4.

If ω_0 and ω_1 form a basis of $H^0(X, K_X - b - c)$, then their ratio $W = \omega_0/\omega_1$ is a rational function on X. The resulting row-reduced-echelon-form normalized pencil was reconstructed over $\mathbf{Q}$ by the Chinese remainder theorem. The least common multiple of the reconstructed coefficient denominators is 91; clearing it gives $J_0, J_1 \in \mathbf{Z}[T, V]$. Their degrees in V are 20 and 21, respectively, and each has degree 6 in T.

4.3. **Reconstruction of the bidegree $(23, 4)$ equation.** Set $W = J_0/J_1$. Eliminating V between $\Phi(T, V)$ and $J_0(T, V) - WJ_1(T, V)$ means extracting the desired factor from

$$\mathrm{Res}_V\big(\Phi(T, V), J_0(T, V) - WJ_1(T, V)\big).$$

We performed this calculation modulo good primes in a stable row-reduced-echelon-form (RREF) normalization, reconstructed the coefficients, and then discarded the numerical evidence in favor of the exact certificate below.

For transparency, the equation used a 575-bit CRT modulus. All 120 coefficients reconstructed and reduced correctly at the independent holdout prime 500015017. The pencil used a 604-bit modulus and passed independent holdout prime 137. These holdouts are evidence about reconstruction, not ingredients in the proof of theorem 1.3; their complete records are `data/f4_crt_reconstruction.json` and `data/pencil_crt_reconstruction.json`. As an auxiliary geometric check, the reduction of P modulo 31 divides the displayed defining resultant; this is verified by `verification/verify_f4_mod31.sing`.

4.4. **Exact certificate and the known degree minimum.** Write

$$P(T, W) = \sum_{j=0}^{23} p_j(T)W^j.$$

Proposition 4.5 (Exact function-field identity)**.** *In $\mathbf{Q}(T)[V]/(\Phi) = \mathbf{Q}(T)[V]/(\widehat{F})$ one has*

$$\sum_{j=0}^{23} p_j(T)J_0^j J_1^{23-j} = 0.$$

Equivalently, $P(T, J_0/J_1) = 0$ in $\mathbf{Q}(X)$.

Proof. This is a polynomial identity modulo $\Phi = F$, equivalently modulo the monic degree-23 relation $\widehat{F}$ over $\mathbf{Q}(T)$. The supplied Singular certificate evaluates the left side by homogeneous Horner reduction and obtains zero. The SageMath certificate independently rebuilds Φ, P, J_0, J_1 from JSON coefficient tables, reads no Singular inputs, and performs the same calculation in its own polynomial arithmetic. A self-contained Magma certificate repeats the quotient-ring identity from the same canonical tables. All computations are exact. $\square$

Proposition 4.6 (Irreducibility)**.** *The polynomial P is primitive and irreducible in $\mathbf{Q}[T, W]$. In particular it is irreducible in $\mathbf{Q}(T)[W]$.*

Proof. The coefficient content is 1. Its reduction modulo 31 retains bidegree $(23, 4)$ and is irreducible in $\mathbf{F}_{31}[T, W]$, as checked independently by Singular, SageMath, and Magma. Gauss's lemma proves the claim. $\square$

Proof of Theorem 1.3. By proposition 4.5, $\mathbf{Q}(T, W)$ is a subfield of $\mathbf{Q}(T, V)$. By proposition 4.6,

$$[\mathbf{Q}(T, W) : \mathbf{Q}(T)] = 23.$$

The source equation also gives $[\mathbf{Q}(T, V) : \mathbf{Q}(T)] = 23$, hence the inclusion is an equality. The regular M_{23} monodromy follows from [1, Proposition 3.5 and Corollary 3.6].

Regard P as a polynomial in T over $\mathbf{Q}(W)$. Since P is irreducible and has positive T-degree, Gauss's lemma and $\deg_T P = 4$ give $[\mathbf{Q}(T, W) : \mathbf{Q}(W)] = 4$. Thus W has degree 4 on X. That this degree is minimal over $\mathbf{Q}$ is the nonsplit-quadric argument already given in [1, Remark 3.4]: every map of degree at most 3 on the geometric genus-4 curve comes from a ruling of its canonical quadric, but neither ruling is defined over $\mathbf{Q}$. Our calculation in proposition 5.21 identifies the quadratic ruling field explicitly. $\square$

5. Supporting arithmetic: the distinguished branch fibre

The minimal equation and the original plane equation describe the same cover, but their coordinates need not display a singular branch fibre in the same way. We first transport the fibre over $T = 0$ exactly between the two models. We then determine its local monodromy, Fano-plane incidence, number fields, and Galois-selected flags, before asking how much of this internal arithmetic can explain the distinguished point of the seven-cover moduli problem.

5.1. **Transport to the minimal-degree coordinate.** The factors in the source coordinate are

$$\begin{aligned} H_7(V) &= V^7 - 312V^6 + 43848V^5 - 3914896V^4 + 241123008V^3 \\ &\quad - 9876755712V^2 + 251799776256V - 2694891036672, \\ R_8(V) &= V^8 + 156V^7 - 1290V^6 - 1217744V^5 - 40611765V^4 \\ &\quad + 2044635024V^3 + 115083889782V^2 + 968032514052V - 6058197515007. \end{aligned}$$

Both are irreducible over $\mathbf{Q}$. The corresponding primitive factors in the minimal-degree coordinate are

$$\begin{aligned} \widetilde{H}_7(W) = {} & 525903119921087741961555W^7 - 838025681952597547830 4W^6 \\ & + 554571400016133612672W^5 - 19338639702208327424W^4 \\ & + 367883711159073536W^3 - 3404298821468160W^2 \\ & + 7203244769280W + 62710185984, \end{aligned}$$

and

$$\begin{aligned}\widetilde{R}_8(W) &= 1055142818078345829104 7W^8 - 2504505235463133310560W^7 \\ &\quad + 2589884858431162696 32W^6 - 1523802213296237772 8W^5 \\ &\quad + 557642314807148032W^4 - 12985270524477440W^3 \\ &\quad + 187639106142208W^2 - 1535652528128W + 5439225856.\end{aligned}$$

Proof of Theorem 1.4. Reconstruct F, P, J_0, J_1 from their canonical integer coefficient tables. Exact factorization over $\mathbf{Q}$ gives the two displayed identities and proves the irreducibility of all four factors. Euclidean reduction gives

$$\gcd(J_1(0,V), H_7R_8) = 1,$$

so the specialized rational function is defined at every geometric point of the fibre.

Let v_7 be the image of V in $K_7 = \mathbf{Q}[V]/(H_7)$ and put

$$w_7 = J_0(0,v_7)/J_1(0,v_7).$$

Exact minimal-polynomial computation gives

$$\operatorname{minpoly}_{\mathbf{Q}}(w_7) = \frac{\widetilde{H}_7}{52590311992108774196155}.$$

In particular w_7 has degree 7 and generates K_7. Substitution therefore defines the first displayed field isomorphism. The identical calculation in $K_8 = \mathbf{Q}[V]/(R_8)$ gives

$$\operatorname{minpoly}_{\mathbf{Q}}\left(\frac{J_0(0,v_8)}{J_1(0,v_8)}\right) = \frac{\widetilde{R}_8}{10551428180783458291047},$$

and proves the second isomorphism. Applying all embeddings into $\overline{\mathbf{Q}}$ shows that the maps on roots are Galois-equivariant and bijective. Every step is reproduced by the certificate described in section 8. □

Remark 5.1. The factor degrees provide a consistency check. The theorem also requires regularity of the rational coordinate throughout the fibre and generation of the full residue fields. The coprimality and minimal-polynomial checks verify these two properties.

5.2. **Local monodromy at the branch point $T = 0$.**

5.2.1. *Group-theoretic notation.* Fix a copy of $M_{23} \subset \mathrm{Sym}(23)$ and elements

$$x \in 2A, \qquad y \in 23A, \qquad xy \in 23A,$$

so that $(x, y, (xy)^{-1})$ is a generating triple with class tuple $(2A, 23A, 23B)$. Set

$$K = O_2(C_{M_{23}}(x)), \qquad H = \mathrm{Fix}(x) \subset \{1, \dots, 23\}.$$

Since x has cycle shape $1^7 2^8$, $|H| = 7$; the fixed letters form a heptad in the Witt design $S(4,7,23)$. One has

$$C_{M_{23}}(x) \cong 2^4{:}\,\mathrm{PSL}_3(2), \qquad |C_{M_{23}}(x)| = 2688, \qquad |K| = 16.$$

For reproducibility, the certificate labeling uses

$$\begin{aligned}x &= (3\,21)(4\,16)(9\,22)(10\,15)(11\,20)(12\,19)(13\,18)(14\,17),\\ y &= (1\,2\,11\,10\,16\,9\,6\,3\,23\,19\,20\,14\,21\,17\,4\,8\,22\,5\,18\,15\,13\,7\,12),\\ r &= (3\,18)(4\,16)(5\,6)(7\,23)(10\,14)(12\,19)(13\,21)(15\,17).\end{aligned}$$

The involution r will compare the displayed triple with its complex conjugate in theorem 5.19. These representatives generate the same Nielsen class as the tuple given in [1, Section 3]; all assertions below are invariant under simultaneous conjugation.

For fixed y let

$$X_y = \{x' \in 2A : x'y \in 23A\}, \qquad |X_y| = 161 = 7 \cdot 23,$$

whose $\langle y\rangle$-translation orbits are the seven Nielsen classes. We call the class containing x the *distinguished class*; it is Nielsen class 6 in the enumeration used below. This is an internal label for the class specified by the displayed permutation x. The exact Hurwitz algebra calculation in section 3 later identifies it intrinsically as the unique degree-one component over K_0.

Put $K_0 = \mathbf{Q}(\sqrt{-23})$. Let $\mathcal{H}_{\mathrm{in}}$ denote the reduced zero-dimensional inner Hurwitz scheme over K_0 for normalized three-point covers with branch-cycle classes $(2A, 23A, 23B)$. Here *inner* means that generating triples are taken modulo simultaneous conjugation by M_{23}; we use the standard Hurwitz-space and Nielsen-class formalism of [6]. Its seven geometric points correspond to the seven $\langle y\rangle$-orbits above. In section 3 we compute a finite K_0-scheme from exact equations and then identify it with $\mathcal{H}_{\mathrm{in}}$ by certified branch cycles.

5.2.2. *Branch points and the fibre over $T = 0$.* The cover is branched over 0 and the two roots of $T^2 + 23$, with local monodromy $(2A, 23A, 23B)$ and total ramification at the order-23 points [1, Section 3]. The valuations and ordinary nodes were computed in lemma 4.1 and Exact computation 4.3. The discriminant separates the branch fibres from the projection singularities.

Proposition 5.2. *The integral model $F \in \mathbf{Z}[T, V]$ satisfies*

$$\mathrm{Res}_V(F, F_V) = uT^8(T^2+23)^{92}G(T)^2, \qquad \mathrm{disc}_V(F) = -uT^8(T^2+23)^{88}G(T)^2, \qquad \deg G = 84,$$

for some $u \in \mathbf{Q}^\times$, where G is squarefree and divides $\gcd(\mathrm{Res}_V(F, F_V), \mathrm{Res}_V(F, F_T))$*, while T does not. Consequently:*

1. *the fibre over $T = 0$ consists of smooth points of the plane model, and $F(0, V) = 23^4 H_7 R_8^2$ gives cycle type $1^7 2^8$, the shape of $2A$;*
2. *the zeros of G account for projection singularities rather than branch points of the normalization; their good reduction modulo 31 is the 84-node scheme of Exact computation 4.3;*
3. *at $T^2 + 23$ the Newton polygon is a single segment of slope $4/23$, so the local extension is totally ramified of degree 23.*

Riemann–Hurwitz gives

$$2g - 2 = -46 + 8 + 22 + 22 = 6, \qquad g = 4.$$

For comparison, direct factorization for the minimal-degree equation gives, up to a nonzero rational constant,

$$\mathrm{disc}_W P = T^8(T^2+23)^{22}H_{62}(T)^2, \qquad \deg H_{62} = 62.$$

The factor T^8 records genuine ramification in both models. The residual square factors and the exponent at $T^2 + 23$ depend on the projection.

5.2.3. *Why the cyclotomic obstruction vanishes.*

Proposition 5.3. *y^λ is conjugate to y in M_{23} if and only if λ is a quadratic residue modulo 23; for the eleven nonsquares it lies in the opposite class $23B$.*

The two order-23 branch points are conjugate over $\mathbf{Q}(\sqrt{-23})$, the quadratic subfield of $\mathbf{Q}(\zeta_{23})$ fixed by the squares. Thus the branch swap matches the nonsquare part of the cyclotomic character. Fried's branch-cycle argument sends (C_1, C_2, C_3) under σ to $(C_1^\lambda, C_2^\lambda, C_3^\lambda)$, with $\lambda = \chi(\sigma)$. For nonsquare λ the tuple becomes $(2A, 23B, 23A)$ while the same σ exchanges the two nonrational branch points. The classical obstruction therefore vanishes, but this permits field of moduli $\mathbf{Q}$ without forcing the fixed point of [1].

5.3. **The Fano plane on the seven unramified points.** The factors H_7 and R_8 were displayed in section 5.1. Their roots are respectively the seven unramified points and eight ramification points above $T = 0$.

Theorem 5.4. $\mathrm{Gal}(H_7) \cong \mathrm{PSL}_3(2)$, *acting on the seven unramified sheets. The* 21 *involutions each fix exactly three sheets. There are seven distinct fixed triples, each arising from three involutions, and these triples form a* 2-$(7, 3, 1)$ *design. Thus the sheets are the points of a Fano plane and the fixed triples are its lines.*

A *flag* means an incident point–line pair in this plane.

Proposition 5.5. *Let* $\alpha_1, \ldots, \alpha_7$ *be the roots of* H_7. *The three-subset-sum resolvent*

$$\mathcal{R}_3(Y) = \prod_{1 \le i < j < k \le 7} \big(Y - (\alpha_i + \alpha_j + \alpha_k)\big)$$

factors over $\mathbf{Q}$ *into irreducible factors of degrees* 7 *and* 28. *The degree-seven factor is*

$$\begin{aligned} L_7(Y) = Y^7 &- 936Y^6 + 379728Y^5 - 86588640Y^4 + 11922182400Y^3 \\ &- 982633068288Y^2 + 44572755885824Y - 859311381083136. \end{aligned}$$

Its roots are the sums along the seven Fano lines. The fields $\mathbf{Q}[V]/(H_7)$ *and* $\mathbf{Q}[Y]/(L_7)$ *are nonisomorphic, but they have the same Galois closure and Dedekind zeta function. Their common field discriminant is*

$$2^4 3^6 23^6 = 1726690609296.$$

Their point and line stabilizers are nonconjugate subgroups of order 24 *with equal permutation characters, the classical Gassmann pair for* $\mathrm{PSL}_3(2)$; *the resulting equality of Dedekind zeta functions is the Gassmann–Perlis criterion* [8].

Corollary 5.6. *The roots of* $\widetilde{H}_7$ *carry the transported Fano-plane point structure. Its line field is defined by* L_7, *and the point and line fields remain nonisomorphic arithmetically equivalent septics.*

Proof. The first isomorphism of theorem 1.4 identifies the two seven-element Galois sets. Fixed triples and incidence are functorial under this identification. □

Proposition 5.7. *The points, lines, and flags admit the following description through* $C_{M_{23}}(x)$. *All* 99 *involutions of* $C_{M_{23}}(x)$ *belong to class* $2A$, *and:*

1. *the fifteen nonidentity elements of* K *fix* H *pointwise. The central element* x *is one conjugacy class and the other fourteen form a second. Writing* $U = K/\langle x \rangle \cong \mathbf{F}_2^3$, *the nonzero elements of* U *identify with the seven lines; the plane is* $\mathbf{P}(U^*)$ *on points and* $\mathbf{P}(U)$ *on lines;*
2. *as an* $\mathbf{F}_2[C_{M_{23}}(x)]$*-module,* K *is indecomposable, with unique minimal nonzero submodule* $\langle x \rangle$. *Each nonzero element of* U *has two preimages in* K, *exchanged by multiplication by* x;
3. *the* 84 *involutions of* $C_{M_{23}}(x) \setminus K$ *form one class and act on* H *as transvections. The map to center and fixed line is a surjection onto the* 21 *flags, with four involutions above each flag. For such* t,

$$C_K(t) = \langle x \rangle \cup \{\textit{the two lifts of the axis}\}, \qquad \mathrm{Fix}(t) \cap H = \mathrm{axis}(t).$$

There is no $C_{M_{23}}(x)$-invariant complement to $\langle x\rangle$ in K. Moreover the outer point–line duality of $\mathrm{PSL}_3(2)$ is not induced by an automorphism of $C_{M_{23}}(x)$ or by a permutation of the 23 sheets normalizing $C_{M_{23}}(x)$:

$$\mathrm{Aut}(C_{M_{23}}(x)) = \mathrm{Inn}(C_{M_{23}}(x)) \quad \text{(order 1344)}, \qquad N_{\mathrm{Sym}(23)}(C_{M_{23}}(x)) = C_{M_{23}}(x).$$

Thus no such symmetry exchanges the point and line Galois sets.

5.4. **Number fields obtained from the fibre.** Put

$$E = \mathrm{Spl}_{\mathbf{Q}}(H_7), \qquad L_0 = \mathrm{Spl}_{\mathbf{Q}}(H_7R_8).$$

We construct a third field M and obtain

$$\mathbf{Q} \subset E \subset L_0 \subset M, \qquad [E:\mathbf{Q}] = 168, \quad [L_0:E] = 8, \quad [M:L_0] = 2.$$

The successive groups over $\mathbf{Q}$ are $\mathrm{PSL}_3(2)$, $2^3{:}\,\mathrm{PSL}_3(2)$, and $2^4{:}\,\mathrm{PSL}_3(2)$. Thus the three stages remember, respectively, the seven unramified points, the eight ramification points, and the choice between the two local branches at each ramification point.

5.4.1. *The splitting field of H_7R_8.* The local decomposition group centralizes x. Since x exchanges the two local branches at a ramification point but fixes its root of R_8, the action on the eight roots factors through $C_{M_{23}}(x)/\langle x\rangle$. The translation subgroup is $U = K/\langle x\rangle \cong \mathbf{F}_2^3$. After choosing an affine origin, the difference of two roots is a nonzero vector of U, hence a Fano line by proposition 5.7; each of the seven nonzero differences occurs for four of the 28 unordered pairs.

Theorem 5.8. *The splitting field of H_7R_8 has Galois group $2^3{:}\,\mathrm{PSL}_3(2)$ of order 1344, acting affinely on the eight roots of R_8. The restriction $\mathrm{Gal}(L_0/\mathbf{Q}) \to \mathrm{Gal}(E/\mathbf{Q})$ has kernel $K/\langle x\rangle \cong (\mathbf{Z}/2)^3$.*

Proof. Decomposition theory at the tame rational place, together with equality of arithmetic and geometric monodromy [1], embeds $\mathrm{Gal}(L_0/\mathbf{Q})$ in $C_{M_{23}}(x)/\langle x\rangle$. Restriction to E is surjective, and its kernel is a submodule of the irreducible three-dimensional module U, hence either trivial or all of U. At $p = 599$, H_7 and L_7 split completely while R_8 factors into four quadratics. Frobenius is therefore trivial on E but a nonzero translation on the roots of R_8, proving that the kernel is U. □

Independently, a direct number-field computation gives $[\mathrm{Spl}_{\mathbf{Q}}(R_8):\mathbf{Q}] = 1344$.

5.4.2. *Affine frames and specialization.* Let A be the eight roots of R_8. An ordered affine frame is a tuple (a_0, a_1, a_2, a_3) whose three differences $a_i - a_0$ form a basis of U. Let

$$\mathcal{F}(A) = \{\text{ordered affine frames of } A\}/U.$$

Proposition 5.9. *$\mathcal{F}(A)$ has 168 elements. The action of $C_{M_{23}}(x)/\langle x\rangle$ descends to a free transitive action of $C_{M_{23}}(x)/K \cong \mathrm{PSL}_3(2)$. Via proposition 5.7, a translation class of frames is equivalently an ordered basis of the Fano line module.*

Proof. There are $8\cdot 7\cdot 6\cdot 4 = 1344$ affine frames. Translation acts freely on the eight origins, leaving 168. The quotient $\mathrm{PSL}_3(2) = \mathrm{GL}_3(2)$ acts freely and transitively on ordered bases of U. □

Let S be the marked geometric generic fibre, based at the tangential basepoint $T \to 0$. For $\sigma \in \mathfrak{G}_{\mathbf{Q}}$, let $d(\sigma)$ be its descent permutation. Since the inertia x has order two, the branch-cycle lemma places $d(\sigma)$ in $C_{M_{23}}(x)$. Write $\rho_8 : \mathfrak{G}_{\mathbf{Q}} \to \mathrm{Sym}(A)$ for the Galois action on the roots of R_8.

Proposition 5.10 (Tame specialization at $T = 0$)**.** *For every $\sigma \in \mathfrak{G}_{\mathbf{Q}}$,*

$$\rho_8(\sigma) = \mathrm{pair}(d(\sigma)),$$

where pair *is the induced action on the eight two-sheet inertia orbits. Consequently the induced action on $\mathcal{F}(A)$ is the one obtained from the generic-fibre descent permutation.*

Proof. Let $R = \mathbf{Q}[T]^h_{(T)}$ and normalize $\operatorname{Spec} R$ in the cover. At a geometric point of ramification index e, tameness and smoothness give the completed local form $T = u^e$ after extracting an eth root of a unit. The e sheets form one inertia orbit and specialize to one point. Hence specialization gives a canonical Galois-equivariant bijection

$$S/\langle x\rangle \xrightarrow{\sim} \{\text{geometric points above } T = 0\}.$$

Restricting to the eight two-element orbits proves the formula. This is the finite-set case of tame specialization [23, Exposé XIII, Section 2.10]. $\square$

5.4.3. *Separating the two branches.* The group $C_{M_{23}}(x)$ is perfect, its center is $\langle x\rangle$, and the central extension

$$1 \longrightarrow \langle x\rangle \longrightarrow C_{M_{23}}(x) \longrightarrow C_{M_{23}}(x)/\langle x\rangle \longrightarrow 1$$

does not split. It acts faithfully and transitively on the sixteen local branches above the eight roots of R_8, with branch stabilizer isomorphic to $\mathrm{PSL}_3(2)$.

Near a root v of R_8, the two branches have expansions

$$V = v \pm \sqrt{e(v)T} + O(T), \qquad e(v) = -\frac{2F_T(0,v)}{F_{VV}(0,v)}.$$

Since $F_{VV}(0,v) = 2 \cdot 23^4 H_7(v) R_8'(v)^2$, the square class of $e(v)$ is represented by

$$c(V) \equiv -F_T(0,V)H_7(V) \pmod{R_8}.$$

Theorem 5.11. *The eight values $c(v)$ represent one nontrivial square class in $L_0^\times/L_0^{\times 2}$. If $M = L_0(\sqrt{c(v)})$ for any root v, then*

$$\mathrm{Gal}(M/\mathbf{Q}) \cong C_{M_{23}}(x) = 2^4{:}\,\mathrm{PSL}_3(2), \qquad [M : L_0] = 2,$$

and the nontrivial element of $\mathrm{Gal}(M/L_0)$ *is x.*

Proof. Tame local theory identifies the splitting field over $\mathbf{Q}((T))$ as $\mathcal{L} = L_0((\sqrt{cT}))$, with Galois group $C_{M_{23}}(x)$. Since $C_{M_{23}}(x)$ is perfect, $\sqrt{T} \notin \mathcal{L}$ and c is not a square in L_0. After adjoining $\sqrt{T}$ the Galois group is $C_{M_{23}}(x) \times \mathbf{Z}/2$. The diagonal involution $(x, -1)$ fixes $\sqrt{c} = \sqrt{cT}/\sqrt{T}$, so taking constant fields gives

$$\mathrm{Gal}(M/\mathbf{Q}) \cong (C_{M_{23}}(x) \times \mathbf{Z}/2)/\langle (x,-1)\rangle \cong C_{M_{23}}(x).$$

Exact arithmetic checks additionally show that $X^2 - dc(V)$ is irreducible over $\mathbf{Q}[V]/(R_8)$ for

$$d \in \{\pm 1, \pm 2, \pm 3, \pm 6, \pm 23, \pm 46, \pm 69, \pm 138\}.$$

At each of fourteen primes below $3 \cdot 10^5$ splitting completely in L_0, the eight values have one quadratic character; at six, beginning with 13049, it is -1, independently certifying nonsquareness. $\square$

The polynomial R_8 has four real roots. Exactly two have $c(v) > 0$, so both local branches above them are real; the other two have $c(v) < 0$, so their branches are exchanged by complex conjugation. This agrees with the permutation r, which fixes the branches above two real ramification points and exchanges the branches above the other two.

Corollary 5.12. *Replacing (H_7, R_8) by $(\widetilde{H}_7, \widetilde{R}_8)$ leaves the fields E, L_0, M, the affine torsor, and the centralizer action unchanged under the transport induced by $W = J_0/J_1$.*

Proof. Theorem 1.4 identifies both residue Galois sets. The first two splitting fields and their functorial actions agree. The final quadratic extension belongs to the normalized cover and separates its local branches, independently of the coordinate labeling the residue points. ☐

5.4.4. *Ramification.*

Theorem 5.13. *Let* $E_8 = \mathbf{Q}[V]/(R_8)$ *and* $E_8' = E_8(\sqrt{c(v)})$*. The fields* $\mathbf{Q}[V]/(H_7)$*,* $\mathbf{Q}[Y]/(L_7)$*, and* E_8 *have discriminants* $2^4 3^6 23^6$*,* $2^4 3^6 23^6$*, and* $2^{16} 3^6 23^6$*. The relative discriminant of* E_8'/E_8 *has norm* $2^8 3^2 23^2$*. Thus* M *is unramified outside* $\{2, 3, 23\}$ *and* M/L_0 *ramifies above all three primes. More precisely:*

1. *at* 23*, the septic fields are totally ramified with* $e = 7$*. Residue inertia is a Singer cycle, transitive on points and lines and fixing one affine origin. The residue decomposition group is* 7:3*; in* M*, inertia is* C_{14} *and decomposition is* $C_2 \times (7{:}3)$*;*
2. *at* 3*, ramification is wild with* $e = 3$ *and lies in the* $\mathrm{PSL}_3(2)$ *quotient;*
3. *at* 2*, the septics are tame with* $e = 3$*, while the octic is wild with* $e = 4$*. The relevant order-*12 *subgroup of* $C_{M_{23}}(x)$ *is* A_4*, with Klein four wild part inside* K *and tame quotient* C_3*.*

Proof. The absolute and relative discriminants are computed from exact maximal orders. At 23, tame local theory gives $\phi\tau\phi^{-1} = \tau^{23} = \tau^2$; since 2 has order three modulo 7, the residue decomposition group is 7:3. The quadratic step adjoins x to inertia. The assertions at 2 and 3 follow from the exact factorization and ramification data for H_7, L_7, R_8 and the subgroup census in the certificates. ☐

5.5. **Flags selected by complex conjugation and Frobenius.**

Theorem 5.14. *Every involution of* $\mathrm{PSL}_3(2)$ *acts as a transvection. It fixes one line pointwise, its axis, and has a distinguished center on that line. The map from involutions to center–axis pairs is a bijection onto the* 21 *flags.*

On points the cycle type is $1^3 2^2$*. Besides the axis, exactly two lines are invariant, each consisting of the center and one transposed pair. The group is transitive on flags; both a flag stabilizer and an involution centralizer have order* 8*.*

The field E *has no real embeddings and* 84 *conjugate pairs of complex embeddings. Each complex place determines the flag of its conjugation involution, giving a four-to-one map from the* 84 *places to the* 21 *flags. The same construction applies at an unramified finite place whose Frobenius is an involution.*

Corollary 5.15. *For complex conjugation, the axis is the set of three real roots of* H_7*. Their sum is a root of* L_7*. The center is the unique real root lying on the other two invariant lines; numerically it is the largest real root, approximately* 102.2684*.*

If p is unramified and H_7 has factorization type $1^3 2^2$ modulo p, then Frobenius is an involution. Such primes have Chebotarev density 1/8.

Theorem 5.16. *Let* N_{28} *be the degree-*28 *nonline factor of* $\mathcal{R}_3$*, and exclude primes dividing* $\mathrm{Res}(L_7, N_{28})$*. At every remaining unramified prime where* H_7 *has type* $1^3 2^2$*, the three roots in* $\mathbf{F}_p$ *sum to a root of* L_7 *and form the axis. The other four roots form two conjugate pairs. Counting, for each fixed root, how many pairs complete a Fano line gives* $(0, 0, 2)$*; the root with count* 2 *is the center. Exact computation checks all* 9,889 *such primes below* 10^6*. Below that bound the only excluded collision is* $p = 22159$*.*

The real axis is defined over $\mathbb{R}$ but not over $\mathbf{Q}$, since $\mathrm{Gal}(E/\mathbf{Q})$ is transitive on the seven lines. At 23 one sees the complementary Singer-cycle picture: after cyclically labeling the seven points, one Fano line is a difference set and the other six are its translates.

5.6. **A first obstruction: geometric reflection.** The branch divisor is preserved by the rational base involution

$$\tau : \mathbf{P}^1_T \longrightarrow \mathbf{P}^1_T, \qquad T \longmapsto -T,$$

which fixes the $2A$ branch point and exchanges the two order-23 branch points. It does not, however, lift to any of the seven covers. The two endpoint inertia generators lie in the distinct classes $23A$ and $23B$. A holomorphic branch swap preserves the orientation of a small loop and therefore preserves its inertia conjugacy class; inversion occurs only for an orientation-reversing comparison such as complex conjugation.

Proposition 5.17 (Geometric reflection obstruction)**.** *The holomorphic involution $\tau : T \mapsto -T$ has no lift to a cover in the Nielsen class $(2A, 23A, 23B)$. In contrast, the orientation-reversing comparison equations*

$$r \in C_{M_{23}}(x), \qquad y^r = z^{-1}, \qquad z^r = y^{-1}$$

have solution counts $(0, 0, 1, 0, 0, 1, 1)$ on the seven normalized triples. Thus IDs $3, 6, 7$ are the classes fixed by the chosen complex-conjugation action.

Proof. For the natural degree-23 action, exact computation gives

$$C_{S_{23}}(M_{23}) = 1, \qquad N_{S_{23}}(M_{23}) = M_{23}.$$

For every normalized triple, y is not conjugate to z in M_{23}, whereas it is conjugate to z^{-1}. Any sheet permutation induced by a holomorphic lift would normalize M_{23} and carry a positive local inertia generator to a conjugate positive generator at the other endpoint. The normalizer identity would place it in M_{23}, contradicting the first class calculation. Complex conjugation reverses the local loop, so its finite comparison uses z^{-1} instead. Exhausting the displayed orientation-reversing equations gives the asserted counts. $\square$

Let

$$Y_x = \{y'' \in 23A : xy'' \in 23A\}.$$

Pairs (x, y'') represent the seven Nielsen classes, and $C_{M_{23}}(x)$ acts on Y_x by conjugation.

Proposition 5.18. *The group M_{23} acts freely on the admissible pairs (x', y') with $x' \in 2A$, $y' \in 23A$, and $x'y' \in 23A$. Moreover*

$$|Y_x| = 18816 = 7 \cdot 2688,$$

and Y_x is the disjoint union of seven free $C_{M_{23}}(x)$-orbits, one per Nielsen class. Equivalently,

$$|Y_x| = |X_y| \frac{|23A|}{|2A|} = 161 \cdot \frac{443520}{3795} = 18816.$$

In these coordinates complex conjugation is

$$\mu : Y_x \longrightarrow Y_x, \qquad \mu(y'') = xy''.$$

Theorem 5.19. *The equation*

$$y''^w = xy'', \qquad w \in C_{M_{23}}(x),$$

has a solution exactly when the corresponding Nielsen class is fixed by complex conjugation. Exactly three classes are fixed. For each, the comparison element w is unique, lies in $C_{M_{23}}(x) \smallsetminus K$, and acts as a transvection on the roots of H_7. For the distinguished class, the unique solution is $w = r$; it satisfies $x^r = x$ and $y^r = xy$. Its center and axis give the real flag of theorem 5.14.

At the chosen real place, complex conjugation exchanges the same two branch points and induces this orientation-reversing permutation on the Nielsen set. The element r acts on the roots of H_7 with type $1^3 2^2$, fixes three Fano lines, and fixes four roots of R_8. Correspondingly, H_7, L_7, R_8 have $3, 3, 4$ real roots. The three fixed classes are the fixed set of this one complex-conjugation element. A $\mathbf{Q}$-defined cubic subscheme would require stability under the full absolute Galois group. The calculation neither establishes such stability nor explains why the distinguished class is globally fixed.

To relate the fibre arithmetic to the fixed-class problem, we now separate what is supplied by the unpointed branch-fibre structures from the effective pointed data used by the quadratic-orientation connector.

5.7. **Why unpointed branch-fibre data do not distinguish the fixed point.** The construction of [1] starts from a seven-element Nielsen class. Its Galois-fixed point produces the rational cover studied here. The Fano plane is attached instead to the seven unramified points in the chosen $2A$ branch fibre of that cover. Thus the results above start from the known rational cover and describe arithmetic internal to it; its descent is an input from [1].

The roles of the structures above may be summarized as follows.

(1) The minimal function W transports the residue fields between the two plane models, while the Fano incidence is intrinsic to the chosen fibre.
(2) The Fano plane is intrinsic to the seven fixed sheets of the chosen $2A$ fibre, but no individual Fano line is Galois invariant over $\mathbf{Q}$.
(3) The affine-frame torsor is compatible with tame specialization from the geometric generic fibre to the fibre over $T = 0$. Distinguishing the rational Nielsen class from the other six requires additional pointed descent data.

The coordinate comparison preserves all Fano and affine residue data, so the construction of §3.5 may be formulated entirely in the minimal-degree model. The branch fibre by itself still does not distinguish the degree-one Hurwitz point. The connector also uses the two marked ends of the characteristic-23 stable map and the effective graph cycles that define the relative half-graded orientation line.

There is also an exact, entirely group-theoretic limit on what unpointed characteristic-23 inertia can determine. Let $I \leq M_{23}$ have order 23 and put $D = N_{M_{23}}(I)$. Direct computation in the supplied permutation model gives $D \cong 23{:}11$. The conjugates of the unpointed inertia subgroup I are therefore indexed by the 40320 points of M_{23}/D. Once a chosen local object is reduced to its unpointed order-23 inertia subgroup, its possible placements are exactly the points of this homogeneous space.

Proposition 5.20 (Unpointed 23-inertia obstruction)**.** *For each of the three complex-conjugation-fixed Nielsen classes, let r be the orientation-reversing comparison of proposition 5.17 and let*

$$A = C_{M_{23}}(x) \cap C_{M_{23}}(r).$$

Then $|A| = 32$, the three ordered pairs (x, r) are simultaneously conjugate in M_{23}, and A acts freely on the 40320 conjugates of I, indexed by M_{23}/D. The involution r divides them into 20160 unordered pairs, on which $A/\langle r\rangle$ acts freely. Consequently there are 1260 orbits of reflection-pairs under $A/\langle r\rangle$ for each fixed Nielsen class. In particular, unpointed order-23 inertia together with (x, r) does not canonically select one pair or distinguish the distinguished class from the other two complex-conjugation-fixed classes.

Proof. Exact computation gives $|A| = 32$ and one simultaneous-conjugacy orbit for the three anchors. Since $|D| = 253$ is odd, the stabilizer in the 2-group A of any coset in M_{23}/D is an intersection with a conjugate of D and is therefore trivial. Thus A acts freely. The element r is central in A and has no fixed coset. An element of A stabilizing one of its unordered two-element orbits either fixes a coset, hence is the identity, or sends it to its r-translate, hence is r. The induced $A/\langle r\rangle$-action is free, and

$$\frac{20160}{|A|/2} = \frac{20160}{16} = 1260.$$

□

Thus a 23-adic argument must augment the two inertia subgroups with local–global comparison data. More precisely, the characteristic-23 Kummer coordinate has two C_{11}-orbits on its nonzero parameters: the squares in $\mathbf{F}_{23}^{\times}$ and their nonsquare coset. On the Fano side, if $F_{\mathrm{fl}} = E^{D_8}$ is a flag field, the two Klein four subgroups $V_4 \subset D_8$ give two quadratic extensions E^{V_4}/F_{fl}. A proposed explanation along these lines would need a Galois-equivariant comparison between these two explicitly defined two-element structures.

There is another natural quadratic datum on the curve, but it does not supply this comparison. Let (A_0, A_1, A_2, A_3) be the RREF-normalized basis of the canonical adjoint space reconstructed from the same 88 coefficient positions used above. Order quadratic monomials as

$$X_0^2, X_0X_1, X_0X_2, X_0X_3, X_1^2, X_1X_2, X_1X_3, X_2^2, X_2X_3, X_3^2.$$

The nonsquare discriminant of this canonical quadric is the obstruction used in [1, Remark 3.4] to prove that degree 4 is minimal over $\mathbf{Q}$. The next proposition refines that argument by identifying the ruling field exactly; that field is also relevant to the local–global comparison pursued here.

Proposition 5.21 (Canonical ruling field)**.** *The unique quadric containing the canonical image, normalized to have coefficient* 1 *at X_3^2, has coefficient vector*

$$\frac{1}{297}\begin{pmatrix} 442260599149123, -18897532752913, -34200754149, 794833324, \\ 168493762065, 745707549, -14145827, 660893, -31345, 297 \end{pmatrix}.$$

If B is twice its symmetric Gram matrix, then

$$\det B = \frac{6444541387164161510 27888}{970299} = 4873\left(\frac{38141181796}{3267}\right)^2, \qquad 4873 = 11 \cdot 443.$$

Consequently the two rulings are governed by

$$\mathbf{Q}(\sqrt{4873}),$$

which is different from the branch-orientation field $\mathbf{Q}(\sqrt{-23})$.

Proof. The four canonical RREF vectors are reconstructed over $\mathbf{Q}$ from 39 good primes with a 604-bit CRT modulus; the ten quadric coefficients use a 141-bit modulus. Both reconstructions agree coefficient by coefficient at the independent holdout prime 137, and the canonical basis reduces to the directly recomputed basis modulo 31. Exact reduction in $\mathbf{Q}(T)[V]/(\widehat{F})$ annihilates the displayed quadratic combination of the ten products A_iA_j; their coefficient matrix has rank 9, proving uniqueness. Exact determinant evaluation gives the displayed square class. The components of the Fano scheme of lines on a smooth quadric surface are defined over its discriminant extension, so the ruling field is $\mathbf{Q}(\sqrt{4873})$. The SageMath and Magma certificates independently repeat the quotient-ring, rank, and determinant calculations. □

The quadratic extension governing the two rulings cannot supply this comparison. Indeed, retain the notation $F_{\mathrm{fl}} = E^{D_8}$ and let E^{V_4}/F_{fl} be either of the two quadratic flag extensions just defined.

If its square class were the product of the square classes obtained from the branch-orientation field $\mathbf{Q}(\sqrt{-23})$ and the ruling field $\mathbf{Q}(\sqrt{4873})$, then

$$E^{V_4} = F_{\mathrm{fl}}\mathbf{Q}(\sqrt{-23 \cdot 4873}).$$

The right side would imply $\mathbf{Q}(\sqrt{-23 \cdot 4873}) \subset E$. This is impossible because $\mathrm{Gal}(E/\mathbf{Q}) \cong \mathrm{PSL}_3(2)$ has no quotient of order two. Thus the ruling field does not provide the required local–global comparison.

The reflection, inertia, and ruling-field calculations determine the reach and the limits of the arithmetic internal to the distinguished cover. They do not by themselves construct the degree-one component of the seven-cover moduli problem. The comparison in section 3.5 succeeds only after adding the pointed logarithmic interval and the two effective graph cycles; it is not a consequence of the unpointed data.

We now leave the branch fibre for an independent global application of the minimal equation.

6. A separate application: arithmetic specializations

At $T = 0$ the specialized polynomial is necessarily reducible and non-squarefree: it records the decomposition and inertia groups at a branch point, not the generic monodromy. Away from the branch locus the same degree-23 equation returns to the full Mathieu group.

We first record the specialization fact that identifies the ambient group.

Lemma 6.1 (Specialization subgroup)**.** *Let $Q(T, x) \in \mathbf{Q}[T, x]$ have splitting field $L/\mathbf{Q}(T)$ with Galois group G. If the leading coefficient and discriminant in x are nonzero at $t \in \mathbf{Q}$, then the splitting field of $Q(t, x)$ has Galois group isomorphic, in its action on the roots, to a subgroup of G.*

Proof. Work over $R = \mathbf{Q}[T]_{(T-t)}$. After dividing by its unit leading coefficient, Q is monic with unit discriminant. The normalization in L is unramified above $T = t$. The residue extension at a chosen prime has Galois group its decomposition group, a subgroup of G. The distinct specialized roots generate that residue extension, proving the claim. ☐

For an integer $t \equiv 2 \pmod{319}$, let c_t be the positive gcd of the coefficients of $P(t, x)$ and set

$$f_t(x) = c_t^{-1} P(t, x) \in \mathbf{Z}[x].$$

Proof of Theorem 1.5. Write $t = 2 + 319n$. Since $319 = 11 \cdot 29$,

$$P(t, x) \equiv P(2, x) \pmod{11}, \qquad P(t, x) \equiv P(2, x) \pmod{29}.$$

The leading coefficient is 5 modulo 11 and 22 modulo 29, so neither prime divides c_t and f_t retains degree 23.

Modulo 29, the monic coefficient vector in increasing order is

$$\begin{aligned}(5, 3, 7, 15, 21, 21, 17, 9, 1, 28, 21, 25, 3, 14, 20, 28,&\\ 5, 27, 24, 16, 9, 20, 18, 1).&\end{aligned}$$

Exact powering gives

$$x^{29^{23}} \equiv x \pmod{\bar{f}_{t,29}}, \qquad \gcd(\bar{f}_{t,29}, x^{29} - x) = 1.$$

Rabin's criterion proves irreducibility because 23 is prime. Thus f_t is irreducible over $\mathbf{Q}$, and Frobenius supplies a 23-cycle.

Modulo 11 one has

$$\bar{f}_{t,11} = g_2 g_7 g_{14},$$

where

$$\begin{aligned} g_2 &= x^2 + 8x + 3, \\ g_7 &= x^7 + 6x^6 + 7x^5 + 6x^4 + 8x^2 + 3x + 1, \\ g_{14} &= x^{14} + 8x^{13} + 6x^{12} + 8x^{11} + 6x^{10} + 6x^9 + 3x^8 + 2x^7 \\ &\quad + 10x^6 + 3x^5 + x^4 + 8x^3 + 10x^2 + 7x + 6. \end{aligned}$$

The factors are distinct and irreducible, so Frobenius supplies an element of cycle type $(2, 7, 14)$ and order 14. Squarefreeness also proves the characteristic-zero discriminant is nonzero, and lemma 6.1 and theorem 1.3 embeds $H_t = \mathrm{Gal}(f_t/\mathbf{Q})$ in M_{23}.

The ATLAS maximal-subgroup orders are [12, 13]

$$443520,\ 40320,\ 40320,\ 20160,\ 7920,\ 5760,\ 253.$$

Only $253 = 23 \cdot 11$ is divisible by 23, and it is odd. A subgroup containing both a 23-cycle and an element of order 14 is therefore not proper. Hence $H_t = M_{23}$. □

Corollary 6.2. *The explicit primitive polynomial $f_2(x) = P(2, x)$ has Galois group M_{23} over $\mathbf{Q}$.*

6.1. **Controlled ramification inside the progression.** Call a family mutually independent if every finite subfamily is linearly disjoint over $\mathbf{Q}$.

Huang et al. already deduce the abstract existence of an infinite mutually independent family from their regular realization [1, Corollary 1.4(d)]. The refinement below chooses such a family inside the explicit progression of theorem 1.5 while forcing a new, controlled inertia prime at each step.

Theorem 6.3. *Among the splitting fields of f_t with $t \equiv 2 \pmod{319}$ there is an infinite mutually independent collection of M_{23}-extensions. More precisely, outside a finite set of primes, if $q \nmid 319$ and*

$$t \equiv 2 \pmod{319}, \qquad v_q(t) = 1,$$

then inertia at q is generated by an element of class $2A$.

Proof. Beckmann's specialization inertia theorem [14, Proposition 4.2], applied at $T = 0$, says that away from finitely many primes, meeting the branch point with multiplicity m produces inertia conjugate to $\langle g^m \rangle$ for local monodromy g. Enlarge this exceptional set to contain 23. If $v_q(t) = 1$, then $T^2 + 23$ is a unit at the specialization, and the inertia group is the class-$2A$ group $\langle g \rangle$.

The Chinese remainder theorem gives

$$t \equiv 2 \pmod{319}, \qquad t \equiv q \pmod{q^2}.$$

Choose each new q outside the bad set and the ramified primes of all previous fields. The new field ramifies at q while the preceding fields do not, so the fields are pairwise distinct.

For mutual independence, suppose $L_1, \ldots, L_r$ are linearly disjoint and $K_r = L_1 \cdots L_r$, with Galois group M_{23}^r. For a new field L, $K_r \cap L$ is Galois over $\mathbf{Q}$. Simplicity of M_{23} makes it either $\mathbf{Q}$ or L. In the latter case a surjection $M_{23}^r \twoheadrightarrow M_{23}$ must factor through one coordinate: the coordinate images are commuting normal subgroups, and simplicity with trivial center permits only one nontrivial image. Then L equals a previous L_i, contradicting distinctness. Hence $K_r \cap L = \mathbf{Q}$. □

Exact computation 6.4 (A second certified field)**.** The integer

$$3830 \equiv 2 \pmod{319}, \qquad 3830 \equiv 5 \pmod{25}$$

lies in the progression and meets $T = 0$ 5-adically with multiplicity one. Over $\mathbf{F}_5$, the repeated part

$$\gcd(F(0,V), F_V(0,V))$$

has degree 8, is squarefree, and is coprime to $F_T(0,V)$. Exact maximal-order computations give

$$v_5(\mathrm{Disc}(\mathbf{Q}[x]/(f_2))) = 0, \qquad v_5(\mathrm{Disc}(\mathbf{Q}[x]/(f_{3830}))) = 8.$$

Thus the $t = 3830$ splitting field ramifies at 5 while the $t = 2$ field does not. The natural degree-23 action is faithful, so the same conclusion holds for their Galois closures: inertia in the closure of the unramified root field fixes every point, hence lies in the core of a point stabilizer, which is trivial.

The local analysis and specialization theorem answer complementary questions. The branch fibre reveals the centralizer and residue arithmetic after a $2A$ collision; the progression gives smooth arithmetic fibres with the full generic group.

7. Open questions

The effective line in theorem 3.15 is deliberately an object of the category of graph-normalization complexes decorated by their integral effective cycles. A natural foundational question is whether it admits an equivalent presentation in a standard hermitian (K)-theoretic or Grothendieck–Witt category of logarithmic correspondences, without first choosing the graph-normalization presentation. Such a presentation would not change the parity calculation, but could make the construction more portable to other Hurwitz problems.

A separate arithmetic classification problem is to classify the $2^4{:}\,\mathrm{PSL}_3(2)$-extensions of $\mathbf{Q}$ unramified outside $\{2, 3, 23\}$ with the local groups and constituent fields of theorem 5.13; in particular, decide whether M is unique.

8. Exact data and certificates

The repository accompanying this paper is

`https://github.com/research-reflexivity/frontier-math-m23-cover-investigation`.

It contains canonical JSON tables for F, P, and the pencil (J_0, J_1). The convention for the minimal-degree equation is

$$\texttt{coefficients\_T\_then\_W[i][j]} = [T^i W^j]P.$$

The polynomial has 120 nonzero coefficients and content 1; its coefficient of $T^4 W^{23}$ is

$$838610528294018704285119511337682389342017079161221371107109332287875.$$

The principal input fingerprints are

```
data/Fint_coefficients_Z.json
27dd1fd0de4f1c350f5a07ead6d5b747d7a8f34e4146ec42fa953f1536a65102
data/optimal_23_4_Z.json
cfb4185b2800744d524d87b3f08f5459cd38d42afb3787c11fa1e2aa3f25ee84
data/optimal_degree4_pencil.json
fdf571e4f47effc527e1726bdd99472be2c5e3390e45133fb3d7a3aab592619d
data/canonical_quadric_Q.json
352b978369353a1cb57a1a3f54edd6b7c024896edf78c38c1601a06f395ac943
data/hurwitz_degree23_branch_candidate.json
cfbce099e3fa867b7c5ba29aa73f02f11d386cf60b00a506d1166d4c65639f62
data/hurwitz_degree23_maps_candidate.json
86cdbd314d01a9123aed0b6890fa4316b4303782b1d69563e020b6ede62a6c2a
```

The single command

```
make verify-all
```

runs the complete open-source suite. The optional target `make verify-magma` repeats the minimal-model identity, irreducibility, canonical-quadric, and unpointed 23-inertia obstruction checks in Magma, together with the twelve-embedding Hurwitz critical-fibre and sextic Galois-closure checks. The recorded environment uses SageMath 10.9 [25], GAP 4.14.0 [26], Singular 4.4.1 [27], and PARI/GP 2.17.4 [28]; the independent Magma records use version 2.29-9 [24]. The opt-in target `make verify-hurwitz-third-fiber-exact` recomputes the longer characteristic-zero resultant certificate. The similarly opt-in target `make verify-hurwitz-connector-a6` reconstructs the formal annulus at the ramified A_6 node used in the quadratic connector. The main targets are:

Result	Certificate	Engine
Newton-polygon valuations, nodes, adjoint pencil	`verify_boundary_valuations.gp`, `verify_nodes_mod31.sing`, `verify_adjoint_mod31.py`	PARI, Singular, Python
Minimal equation and function-field identity	`verify_optimal_23_4.py`, `verify_sage.py`, `verify_optimal_23_4.m`	Singular, SageMath, Magma
Specialization progression and ramification	`verify_progression_319.*`, `verify_ramified_t3830.*`	Python, PARI
Transport of the fibre over $T = 0$	`verify_special_fibre_bridge.py`	SageMath
Branch cycles and Fano incidence	`certify_branch_cycle_description.*`, `certify_special_fibre_fano_*`	GAP, PARI
Affine field tower and final quadratic layer	`certify_special_fibre_field_tower.*`, `certify_vanishing_cycle_class.*`	GAP, PARI
Ramification, flags, and complex-conjugation comparison	`certify_special_fibre_ramification.*`, `certify_local_flag_places.*`, `certify_real_frame_cocycle.*`	GAP, PARI
Geometric reflection obstruction and complex-conjugation comparison	`certify_geometric_reflection_*`	GAP, Magma
Canonical quadric and ruling field	`verify_canonical_quadric.py`, `verify_canonical_quadric.m`	SageMath, Magma
Unpointed 23-inertia obstruction	`certify_gauss_prolongation_obstruction.g`, `certify_gauss_prolongation_obstruction.m`	GAP, Magma
Coordinate algebra of the Hurwitz scheme, canonical curves, and marked points	`verify_hurwitz_*_candidate.py`	SageMath
Uniform atlas cover and truncation-error bounds	`verify_hurwitz_tail_geometry.py`, `verify_hurwitz_tail_summary.py`	Arb, Acb, Python
Degree-23 ratios, third branch value, and complete passport	`verify_hurwitz_degree23_*.py`, `verify_hurwitz_degree23_third_fiber.sage`, `verify_hurwitz_degree23_branch.m`, `verify_hurwitz_degree23_geometry.m`	SageMath, Magma
Exact degree-23 eliminant and all seven branch-cycle triples	`verify_hurwitz_monodromy_eliminant_candidate.py`, `verify_hurwitz_branch_cycle_summary.py`	SageMath, Arb

Result	Certificate	Engine
Relative-transporter invariants and component-idempotent descent	`certify_hurwitz_relative_transporter.g`, `verify_hurwitz_relative_transporter.py`	GAP, SageMath
Galois closure and six-point permutation action	`verify_hurwitz_galois_closure.*`	SageMath, PARI, Magma
Local decomposition group and pointed reductions at 23	`verify_hurwitz_local_23.py`, `verify_hurwitz_pointed_23.py`	SageMath
Finite and local ingredients for the effective quadratic connector	`notes/certify_ade_gluing_marker.py`, `notes/certify_wild_parameter_orientation.py`, `notes/audit_lifted_trace_fano_affine_incidence.g`, `notes/audit_fano_affine_odd_fixed_point_lemma.g`, `notes/audit_raw_lifted_trace_node_boundary.g`, `notes/audit_returned_normalizer_trace_parity.g`, `notes/explore_tagged_tame_boundary.g`, `notes/certify_pinched_tag_finite_identities.py`, `notes/audit_pointed_relative_bockstein.py`, `notes/audit_pointed_relative_bockstein.m`, `notes/audit_log_quadratic_orientation_line.py`, `notes/audit_log_quadratic_orientation_line.m`	GAP, Python, Magma

Several certificate filenames predate the terminology adopted in this paper and retain the string `special_fibre`. In those filenames it refers to the characteristic-zero fibre over $T = 0$, not to the closed fibre of a model over a local base.

For the coordinate-transport certificate, SageMath reconstructs the fibres over $T = 0$ in both plane models, checks the factor multiplicities and scalar identities, proves $\gcd(J_1(0, V), H_7R_8) = 1$, and computes the minimal polynomial of J_0/J_1 in each residue field. Thus the certificate verifies the coordinate transport coefficient by coefficient, including regularity and residue-field generation. The largest computations in the Fano suite are the direct degree-1344 splitting-field calculation for R_8, the comparison of the possible septic/octic Frobenius types, the 9,889 transvection primes below 10^6, and the square-class tests at fourteen primes splitting completely in L_0.

9. Scope, provenance, and disclosure

The starting cover is not constructed in this paper. The regular M_{23} realization, its source equation, branch cycles, and the numerical computation of the seven complex covers and their Galois-fixed Nielsen class are due to Huang, Jackson, Lee, Poonen, Pries, and Zhang [1]. Their accompanying repository supplies the defining data used here [2].

The contributions of the present paper fall into four groups.

(1) We reconstruct the coordinate algebra of the inner Hurwitz scheme, its seven canonical curves and degree-23 maps, their complete passports, and their branch cycles, and determine the S_6 Galois action on the degree-six component.
(2) We determine the local decomposition and pointed reductions of that scheme at 23, construct the idempotent defined by the singular-point coordinate directly from those reductions, determine the descent behavior of the relative-transporter invariants, and compare their component idempotents in the exact reconstructed algebra. We also construct

the effective logarithmic quadratic-orientation line and obtain a second comparison by specialization, while proving that the two natural un-enriched nearby-cycle models cannot carry it.

(3) On the distinguished cover, we explicitly construct the degree-4 function W whose existence and minimality were established in [1, Remark 3.4], its bidegree-$(23, 4)$ equation, and exact certificates that it defines the same function field. We determine the arithmetic of the fibre over $T = 0$, including its Fano incidence, affine torsor, field tower, ramification, and obstruction theory, and transport this structure to the minimal coordinate.

(4) We prove the explicit uniform specialization theorem and controlled ramification inside its progression, and construct within that progression an infinite mutually independent family. The abstract existence of such families is already [1, Corollary 1.4(d)].

We do not reprove that the fixed Nielsen class is rational; that is a theorem of [1]. The idempotent defined by the singular-point coordinate gives an intrinsic characteristic-23 criterion separating the degree-one and degree-six components of the reconstructed algebra. The comparison theorem proves its agreement with the relative-transporter augmentation by using the exact branch-cycle identification of all seven reconstructed maps. The effective logarithmic orientation line of theorem 3.15 and corollary 3.16 gives a second proof from the pointed characteristic-23 geometry. It is not an ordinary nearby-cycle determinant: the nearby-cycle stalk retains the sheet labels and the coefficient Bockstein vanishes in the natural un-enriched models.

The third branch fibre of the six conjugate maps is verified exactly in characteristic zero and independently in all twelve residue embeddings of a completely split prime. This distinction matters: numerical and lattice reconstruction discover the coefficients, while the subsequent exact elimination and branch-cycle certificates identify the covers.

The discovery and drafting process used substantial AI assistance. Every new computational assertion is covered by an exact certificate described in section 8.

Acknowledgements. This work analyzes the construction of Huang, Jackson, Lee, Poonen, Pries, and Zhang and depends on their making the defining data publicly available in the repository cited as [2].

References

[1] X. Huang, B. Jackson, K.-H. Lee, B. Poonen, R. Pries, and S. Zhang, *The Mathieu group M_{23} is a Galois group over* **Q**, arXiv:2608.08538v1 (2026), `https://arxiv.org/abs/2608.08538`.
[2] S. Zhang et al., *m23isgalois*, data repository, `https://github.com/shaowuz/m23isgalois`.
[3] F. Häfner, *Braid orbits and the Mathieu group M_{23} as Galois group*, arXiv:2202.08222 (2022).
[4] M. Klug, M. Musty, S. Schiavone, and J. Voight, *Numerical calculation of three-point branched covers of the projective line*, LMS J. Comput. Math. **17** (2014), 379–430.
[5] M. Musty, S. Schiavone, J. Sijsling, and J. Voight, *A database of Belyi maps*, Open Book Ser. **2** (2019), 375–392.
[6] M. D. Fried and H. Völklein, *The inverse Galois problem and rational points on moduli spaces*, Math. Ann. **290** (1991), 771–800.
[7] M. Rosenlicht, *Equivalence relations on algebraic curves*, Ann. of Math. (2) **56** (1952), 169–191.
[8] R. Perlis, *On the equation $\zeta_K(s) = \zeta_{K'}(s)$*, J. Number Theory **9** (1977), 342–360.
[9] A. K. Lenstra, H. W. Lenstra, Jr., and L. Lovász, *Factoring polynomials with rational coefficients*, Math. Ann. **261** (1982), 515–534.
[10] F. Johansson, *Arb: Efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), 1281–1292.
[11] G.-M. Greuel, C. Lossen, and E. Shustin, *Introduction to singularities and deformations*, 2nd ed., Springer Monographs in Mathematics, Springer, Cham, 2025.
[12] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker, and R. A. Wilson, *Atlas of finite groups*, Clarendon Press, Oxford, 1985.

[13] The ATLAS of Finite Group Representations, *Mathieu group M_{23}: maximal subgroups*, `https://brauer.maths.qmul.ac.uk/Atlas/v3/spor/M23/`.

[14] S. Beckmann, *On extensions of number fields obtained by specializing branched coverings*, J. Reine Angew. Math. **419** (1991), 27–54.

[15] P. Dèbes and N. Ghazi, *Galois covers and the Hilbert–Grunwald property*, Ann. Inst. Fourier (Grenoble) **62** (2012), no. 3, 989–1013.

[16] F. Legrand, *Specialization results and ramification conditions*, Israel J. Math. **214** (2016), no. 2, 621–650.

[17] F. Legrand, *On finite embedding problems with abelian kernels*, J. Algebra **594** (2022), 51–73.

[18] D. Ferrand, *Conducteur, descente et pincement*, Bull. Soc. Math. France **131** (2003), no. 4, 553–585, `https://doi.org/10.24033/bsmf.2455`.

[19] F. Knudsen and D. Mumford, *The projectivity of the moduli space of stable curves. I: Preliminaries on "det" and "Div"*, Math. Scand. **39** (1976), 19–55, `https://doi.org/10.7146/math.scand.a-11642`.

[20] C. Nakayama, *Nearby cycles for log smooth families*, Compositio Math. **112** (1998), no. 1, 45–75, `https://doi.org/10.1023/A:1000327225021`.

[21] S. Wewers, *Three point covers with bad reduction*, J. Amer. Math. Soc. **16** (2003), no. 4, 991–1032, `https://doi.org/10.1090/S0894-0347-03-00435-1`.

[22] Q. Lu and W. Zheng, *Categorical traces and a relative Lefschetz–Verdier formula*, Forum Math. Sigma **10** (2022), e10, `https://doi.org/10.1017/fms.2022.2`.

[23] A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental (SGA 1)*, Documents Mathématiques **3**, Société Mathématique de France, 2003; arXiv:math/0206203.

[24] W. Bosma, J. Cannon, and C. Playoust, *The Magma algebra system I: The user language*, J. Symbolic Comput. **24** (1997), 235–265.

[25] The Sage Developers, *SageMath, the Sage mathematics software system*, version 10.9, 2026, `https://www.sagemath.org`.

[26] The GAP Group, *GAP—Groups, Algorithms, and Programming*, version 4.14.0, 2024, `https://www.gap-system.org`.

[27] W. Decker, G.-M. Greuel, G. Pfister, and H. Schönemann, *Singular 4-4-1—A computer algebra system for polynomial computations*, 2025, `https://www.singular.uni-kl.de`.

[28] The PARI Group, *PARI/GP*, version 2.17.4, Université de Bordeaux, 2026, `https://pari.math.u-bordeaux.fr`.

Applied Science Team, Reflexivity, 589 5th Avenue, Suite 606A, New York, NY 10017
Email address: `francois@reflexivity.com`
URL: `https://www.reflexivity.com`